

Measuring Media Bias Toward Reform Prosecutors: A Multi-Method NLP Analysis of Bay Area News Coverage, 2019–2024

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Abstract

Media coverage is the primary channel through which the public evaluates prosecutorial policy, yet whether the press covers reform-oriented prosecutors differently from their traditional counterparts remains untested at scale. I analyze 136,313 news articles (2019–2024) drawn from 19 Bay Area outlets and two national outlets that cover these prosecutors, using a multi-method NLP pipeline that separately measures emotional tone, evaluative stance, thematic emphasis, narrative framing, and source ecology across five district attorneys spanning the progressive–traditional spectrum. Media bias toward prosecutors is not a unitary phenomenon: its dimensions diverge sharply. Emotional tone surrounding prosecutor mentions is statistically equivalent across ideological groups (per-method equivalence tests bound both sentiment effects within $|d| < 0.2$, $p < .001$), yet evaluative stance classification reveals substantially more critical framing of progressive prosecutors, an effect confirmed by within-county comparisons and robust to regression controls. The pooled cross-sectional composite contrast ($d = -0.151$) is roughly twice the within-county estimate and halves when the sole San Mateo County prosecutor is dropped from the traditional baseline, so the designs that avoid cross-county comparison converge on a smaller effect ($d \approx -0.07$) than the pooled con-

trast implies. When critical themes are explicitly attributed to a named prosecutor, the raw differential is larger still ($d = -0.418$; throughout, a negative d indicates coverage less favorable to the progressive prosecutor), but it is substantially event-driven: excluding articles that invoke recall campaigns or rising-crime claims—events that differed between the groups, because both progressive prosecutors faced actual recall efforts and no traditional prosecutor did—reverses its sign ($d = +0.088$). Progressive prosecutors receive accountability framing at more than twice the rate of traditional counterparts (40% vs. 19%), who are more often covered through human-interest narratives; framing contrasts are reported on the model-classified subset, because pooled frame scores mix a zero-shot classifier with a keyword fallback on incommensurable scales and are not interpretable. LLM-based structural extraction reveals a corresponding asymmetry in source ecology: progressive coverage draws disproportionately on advocacy, expert, and politician voices, while traditional coverage relies more heavily on the prosecutor’s own statements. These differentials are event-driven rather than reflecting a stable editorial disposition. The press covers reform prosecutors differently in kind, not in degree: through systematic differences in evaluative framing, sourcing, and narrative structure that are invisible to tonal analysis alone.

1 Introduction

The progressive prosecution movement represents one of the most significant developments in American criminal justice over the past decade. Beginning with the election of reform-minded district attorneys in major cities (including Larry Krasner in Philadelphia (2017), Kim Foxx in Chicago (2016), and Chesa Boudin in San Francisco (2019)) a new generation of prosecutors has sought to reduce mass incarceration, curtail cash bail, decline to prosecute low-level offenses, and hold law enforcement accountable (Bellin, 2019; Sklansky, 2017). These prosecutors have faced intense political opposition, including recall campaigns, legislative challenges, and sustained media scrutiny (Bazon, 2019). Goldstein (2024) characterizes this resistance as “toplash”: elite-driven opposition from wealthy donors, police unions, and political establishments that is distinct from grassroots backlash and specifically targets prosecutors’ reform agendas through media campaigns and institutional pressure.

Media coverage plays a crucial role in shaping public attitudes toward criminal justice reform. How prosecutors are portrayed in the press, whether through sympathetic human-interest narratives or critical accountability frames, can influence public support for reform policies and, ultimately, electoral outcomes (Iyengar, 1991; Entman, 1993). The recall of San Francisco District Attorney Chesa Boudin in June 2022, widely attributed to perceptions of rising crime and prosecutorial leniency, underscores the political salience of media framing in this domain (Yogev, 2026).

Despite the importance of media coverage in the politics of prosecution, there has been little systematic analysis of whether progressive prosecutors receive measurably different treatment from the press compared to their traditional counterparts. Existing scholarship on media bias in criminal justice has focused primarily on race and crime (Dixon and Linz, 2000), sentencing coverage (Surette, 2015), and general ideological slant at the outlet level (Grosseclose and Milyo, 2005). The prosecutor-specific question, whether the press covers reform-oriented officeholders more negatively and if so through what mechanisms, remains largely unexamined. The most systematic prior effort, a pilot study of prosecutor coverage across four states, found that the median incumbent

prosecutor appeared in only 24 articles during an entire election year, and 89 percent of that coverage contained no information about prosecutors’ decisions or policies (Hessick and Thornburg, 2023).

This study addresses that gap. I analyze 136,313 news articles from 21 publications over a five-year period (2019–2024)—19 Bay Area outlets plus two national outlets, *The New York Times* and *Politico*, that cover these prosecutors—comparing coverage of five district attorneys who span the progressive–traditional spectrum. The San Francisco Bay Area provides an unusually strong research design: multiple counties with sequential tenures of progressive and traditional prosecutors in the same jurisdiction allow within-county comparisons that control for local media markets, crime conditions, and editorial cultures.

I contribute to the literature in three ways. First, I develop a multi-method NLP pipeline that goes beyond simple sentiment analysis to measure aspect-based sentiment (tone specifically toward the prosecutor, not the article generally), zero-shot stance classification (whether the article’s framing criticizes or defends the prosecutor), and media frame detection (the narrative lens through which prosecutors are presented). Second, I provide the first large-scale empirical test of differential media treatment of prosecutors by ideological orientation. Third, I demonstrate that framing differences, specifically the disproportionate use of accountability and reform frames for progressive prosecutors, may be more consequential than aggregate tone differences.

2 Literature Review

2.1 Media Framing and Criminal Justice

The study of media framing in criminal justice draws on Entman’s (1993) definition of framing as selecting and emphasizing particular aspects of perceived reality to promote specific interpretations. In criminal justice contexts, framing choices—whether to emphasize individual responsibility versus systemic factors, crime trends versus policy outcomes, victims versus defendants—

shape public understanding of complex policy questions (Iyengar, 1991; Gross, 2008). Tversky and Kahneman (1981) foundational work on framing effects demonstrates that logically equivalent information presented in different frames produces systematically different judgments, a finding repeatedly confirmed in criminal justice settings (Slovic et al., 2000).

Research on crime news specifically has documented a persistent “mean world” effect (Gerbner et al., 2002): local news overrepresents violent crime relative to its actual incidence, contributing to public fear and support for punitive policies. Dixon and Linz (2000) demonstrate that racial minorities are overrepresented as perpetrators and underrepresented as victims in local television news, while Gilliam and Iyengar (2000) show that implicit racial cues in crime coverage activate punitive policy preferences among white viewers. The asymmetry extends to victim portrayal: white victims are significantly more likely than Black victims to receive sympathetic coverage featuring personal photographs and family context (Global Strategy Group and Equal Justice Initiative, 2021). These media-constructed racial associations drive support for punitive policies independently of actual crime rates (Ghandnoosh, 2014), a dynamic directly relevant to reform prosecutors who frame their agendas around addressing racial disparities in prosecution. This body of work establishes that media coverage of criminal justice is neither neutral nor proportional to underlying reality.

These patterns reflect structural features of the media industry rather than individual editorial choices. Beale (2006) argues that commercial pressures determine the news media’s contemporary treatment of crime: sensationalized coverage is cost-effective and audience-attracting, creating persistent overrepresentation of violent crime. Market incentives also generate structural dependencies on law enforcement as readily available, authoritative sources, a reliance that shapes the narrative framework within which prosecutors are evaluated (Beale, 2006; Surette, 2015). These media dynamics have measurable downstream consequences: Ash and Poyker (2024) provide causal evidence that exposure to conservative crime coverage increases sentencing harshness among elected judges, while Romer et al. (2003) confirm that local television news viewing increases fear of

crime independent of actual crime rates.

2.2 Prosecutorial Politics and Public Perception

The rise of the progressive prosecution movement has generated a growing scholarly literature on prosecutorial discretion, electoral accountability, and political backlash (Pfaff, 2017; Sklansky, 2017; Bellin, 2019). Wright (2009) provides a framework for understanding prosecutorial elections as mechanisms of democratic accountability, while Davis (2007) documents the broad discretionary powers that make the prosecutor’s office a site of significant policy variation. Bazelon (2019) chronicles the political dynamics surrounding progressive prosecutors, noting that opponents have mobilized law enforcement unions, victims’ rights organizations, and media campaigns to challenge reform agendas. Police unions occupy a dual position in this dynamic: they serve as primary media sources for crime stories while simultaneously funding opposition campaigns against reform prosecutors and lobbying for legislation limiting prosecutorial discretion (Goldstein, 2024). This structural asymmetry means that the most frequently quoted institutional actors in crime coverage are also the most organized opponents of prosecutorial reform. Recent reviews document the scope of this opposition, including legislative challenges across multiple states and coordinated recall campaigns (Mitchell and Petersen, 2025).

However, empirical analysis of media coverage of individual prosecutors remains scarce. Despite prosecutors’ central role in criminal justice, they are structurally under-covered: a UNC pilot study found that prosecutors receive minimal media attention, with coverage rarely mentioning their elected status even during election years (Hessick and Thornburg, 2023). This coverage deficit is stark relative to law enforcement; in the crime-relevant corpus analyzed here, 83 percent of articles mention police or another law enforcement actor, compared with 16 percent that mention the district attorney.¹ The scarce attention that prosecutors do receive is thus particularly high-stakes,

¹Computed over the 107,713 articles passing the relevance filter (output/03_all_relevant_attributed.parquet), searching headline and body jointly with word-boundary-anchored matching. “Mentions police” is any of *police*, *officer(s)*, *sheriff*, *deputy/deputies*, *law enforcement*, *detective(s)*, or *highway patrol*: 89,471 articles, 83.1%. “Mentions the district attorney” is any of *district attorney*, *D.A.*, or *DA’s office*: 16,717 articles, 15.5%. The gap is sensitive to

as it constitutes the primary channel through which the public evaluates prosecutorial policy. Most studies of media bias in politics focus on candidate coverage in electoral contexts (D'Alessio and Allen, 2000; Puglisi and Snyder, 2011) or ideological slant at the outlet level (Groseclose and Milyo, 2005; Gentzkow and Shapiro, 2010). The prosecutor-specific question is distinct because prosecutors are both political figures and quasi-judicial actors, operating in a policy domain, criminal justice, where media coverage has documented and substantial effects on public attitudes.

2.3 Computational Text Analysis in Political Communication

Recent advances in natural language processing have expanded the toolkit available for media bias measurement. Transformer-based models, including BERT (Devlin et al., 2019) and its derivatives, have achieved state-of-the-art performance on sentiment analysis, stance detection, and text classification tasks (Liu et al., 2019). Zero-shot classification approaches, which leverage models trained on natural language inference to classify text without task-specific training data, are particularly valuable in political communication research where labeled training corpora are scarce (Yin et al., 2019).

Applications of NLP to media bias detection include Hamborg et al. (2019), who develop a framework for identifying framing bias through word choice and emphasis, and Fan et al. (2019), who use BERT-based models to detect informational and lexical bias in news articles. Card et al. (2015) apply latent frame analysis to immigration coverage, demonstrating that computational methods can reliably detect framing patterns at scale. This study builds on these approaches by integrating multiple NLP measurements (sentiment analysis, stance detection, thematic emphasis, and frame classification) to capture distinct but related dimensions of media treatment.

how broadly law enforcement is defined—restricting to *police*, *sheriff*, *deputies*, *law enforcement*, and *highway patrol* alone gives 79.4%—but not to any specification that reverses the comparison. Broadening the prosecutor term to include *prosecutor(s)* raises that side to 24.7%, still roughly a third of the police rate.

3 Data and Methods

The empirical strategy proceeds from sample construction to complementary measurement layers and then to comparative inference. I first construct a prosecutor-attributed corpus from the full news archive through relevance filtering and attribution/disambiguation. I then estimate four directional prosecutor-focused scores (Methods A–D) on a common signed scale and combine them into a composite index for a parsimonious overall test. Framing is estimated separately because it is a categorical narrative-lens outcome rather than a signed intensity measure. Prosecutor-attributed theme detection is also separate because it measures explicit prosecutor–theme linkage rather than local thematic salience alone. Finally, LLM structural extraction captures article architecture (who is quoted, what claims are made, and how causality is asserted), which complements score-based NLP by measuring content structure directly. Inference combines group and paired comparisons, controlled regressions, and segmented interrupted time-series models with quarterly heterogeneity summaries.

3.1 Data

The analysis draws on a corpus of 136,313 news articles published between January 2019 and December 2024, collected from LexisNexis Academic across 21 publications. Nineteen are Bay Area outlets: regional newspapers and their digital siblings (sfgate.com, 40,888 articles; *San Francisco Chronicle*, 13,680; *East Bay Times*, 12,264; *San Mateo Daily Journal*, 5,432; sfist.com, 3,835; sfexaminer.com, 3,011), television station web content (kron4.com, 14,863; sanfrancisco.cbslocal.com, 9,572; nbcbayarea.com, 9,276; abc7news.com, 8,504; ktvu.com, 264), public media (kqed.org, 2,836), and digital-native local news sources (missionlocal.org, 3,004; berkeleyside.org, 1,390; sfbayview.com, 1,132; oaklandside.org, 913; alamedapost.com, 169; sfstandard.com, 52; independentnews.com, 5).

The remaining two publications are not Bay Area outlets, and this should be stated plainly rather than glossed: nytimes.com (3,884 articles) and politico.com (1,339) are national publications that

enter the corpus because they cover these prosecutors, particularly the Boudin recall. Together they supply 5,223 articles, 3.8% of the full corpus, and 395 of the 12,827 prosecutor-attributed articles analyzed below (159 from nytimes.com, 236 from politico.com, 3.1% of the analysis sample). The corpus is therefore predominantly but not exclusively local. The national component is small enough that it cannot drive the aggregate contrasts, but it is not neutrally distributed: 306 of those 395 attributed articles (77%) concern a progressive prosecutor, against 51% in the analysis sample overall, and 292 concern Boudin specifically. National attention followed the recall, which is exactly what one would expect and exactly why the two outlets are disclosed here rather than folded silently into a “Bay Area” count (see Section 5.2).

One reconciliation is worth stating here because two publication counts appear in this paper. The full corpus spans 21 publications, but only 20 contribute at least one prosecutor-attributed article to the analysis sample; independentnews.com, with five articles in the full corpus, contributes none. The publication-cluster bootstrap described in Section 3.7 accordingly resamples 20 outlets, not 21. The temporal range spans the full tenure of Chesa Boudin as San Francisco District Attorney (2020–2022), the subsequent tenure of his replacement Brooke Jenkins (2022–present), the transition from Nancy O’Malley to Pamela Price in Alameda County (2023), and the ongoing tenure of Steve Wagstaffe in San Mateo County.

3.2 Article Filtering and Prosecutor Attribution

Not all articles in the corpus concern crime and criminal justice. I apply a relevance filter requiring at least two crime/justice terms (prosecutor, arrest, sentencing, felony, etc.), which retains 107,713 articles; a zero-shot classifier (BART-MNLI; Lewis et al. 2020) serves as a secondary relevance screen for additional validation checks. Prosecutor attribution then uses regex-based name matching across known variants (e.g., “Boudin,” “Chesa Boudin,” “DA Boudin”) with contextual disambiguation for “Price” (counted as Pamela Price only within 50 words of terms such as “DA,” “prosecutor,” “Alameda,” or “Pamela”). Articles are assigned a primary prosecutor by mention

frequency, with headline mentions weighted at $3\times$. A further disambiguation step applies a time-symmetric confirmation rule to the ambiguous surnames “Jenkins” and “Price”: in every period, a surname match is accepted only if the prosecutor’s first name appears somewhere in the article or a prosecutorial title is immediately adjacent to the surname (e.g., “District Attorney Jenkins”), eliminating false positives from unrelated individuals who share those surnames and from non-name uses of “price.” An earlier version of this filter required first-name confirmation only *before* each prosecutor took office; symmetrizing it removed 126 post-tenure false positives (102 for Jenkins, 24 for Price), all concerning other people named Jenkins or Price, and reassigned 26 articles to the correct prosecutor. Because these false positives accumulated exclusively after the tenure start dates, they constituted a time-asymmetric contamination located precisely on the interrupted time-series breakpoints analyzed below; that a mundane matching rule could produce so sharply patterned an artifact is itself evidence of the rule’s importance. The superseded asymmetric attribution is archived with the replication outputs (output/archive_03_asymmetric/) for comparison. When no named prosecutor appears but the article refers to “district attorney”/“prosecutor,” attribution falls back to county-by-date officeholder assignment; these cases are flagged and evaluated in sensitivity analyses. This process yields $n = 12,827$ articles attributed to five prosecutors (Table 1). Importantly, the relevance filter uses generic crime and justice vocabulary (“prosecutor,” “arrest,” “felony”) that is orthogonal to the outcome-specific theme dictionaries applied later (“soft-on-crime,” “recall campaign,” “public-safety-failure”); the filtering step therefore does not select articles on the basis of the thematic variables subsequently measured.

Table 1: Prosecutor sample characteristics.

Prosecutor	County	Ideology	Tenure	n
Chesa Boudin	San Francisco	Progressive	2020–2022	5,783
Brooke Jenkins	San Francisco	Traditional	2022–present	3,097
Nancy O’Malley	Alameda	Traditional	2009–2023	1,497
Pamela Price	Alameda	Progressive	2023–2024	802
Steve Wagstaffe	San Mateo	Traditional	2010–present	1,648
<i>Progressive total</i>				6,585
<i>Traditional total</i>				6,242

3.3 Multi-Method Bias Detection

I estimate four prosecutor-focused article-level scores (Methods A–D) and combine them into a weighted composite used for the paper’s primary directional comparisons. These methods are grouped because each produces a signed directional measure on a common $[-1, +1]$ scale. Worked examples of what each method flags in actual corpus articles are collected in Appendix D.

Method A: Aspect-based sentiment analysis (weight: 0.35). For each prosecutor mention in an article, I extract a three-sentence context window centered on the mention and compute sentiment polarity using a fine-tuned RoBERTa model (cardiffnlp/twitter-roberta-base-sentiment-latest; Loureiro et al. 2022). The article-level score averages across mention windows, measuring sentiment *toward the prosecutor* rather than overall article tone—a critical distinction in crime reporting, where subject matter is inherently negative.

Method B: Zero-shot stance classification (weight: 0.30). For paragraphs containing prosecutor mentions, I classify stance using BART-MNLI (Lewis et al., 2020) with three labels: criticism, defense/support, and neutral reporting about the prosecutor. The article-level score is the balance of supportive minus critical paragraph probabilities.

Method C: Enhanced keyword analysis (weight: 0.20). I apply curated theme dictionaries (crime-rising, soft-on-crime, recall, case dismissal, releasing criminals, public safety failure, victim neglect, police conflict, and office dysfunction) within three-sentence windows around prosecutor mentions. A negation check in the preceding words reduces false positives. These dictionaries operationalize the anti-prosecutor framing patterns identified in prior qualitative work on prosecutorial media coverage (Bazelon, 2019; Goldstein, 2024).

Method D: Document-level sentiment baseline (weight: 0.15). Whole-article sentiment serves as a control, capturing general negativity in crime reporting that is not necessarily prosecutor-specific.

Composite construction (A–D only). The composite score is the weighted average of Methods A–D (0.35, 0.30, 0.20, 0.15) on a common $[-1, +1]$ direction where more negative values indicate more negative treatment. If one or more method scores are missing, weights are renormalized over available methods. The weights follow a precision–generality hierarchy: the two prosecutor-targeted measures (aspect sentiment, 0.35; stance, 0.30) are prioritized over the two text-level measures (keywords, 0.20; document sentiment, 0.15). This scheme is empirically conservative: it assigns the highest weight to the method with the weakest effect (aspect sentiment, $d = 0.047$) and lower weights to the methods with larger effects (stance $d = -0.337$; keywords $d = -0.203$), mechanically attenuating the composite. Eight alternative weighting specifications (equal, inverse, effect-proportional, evaluative-only, and four leave-one-out variants) show that the composite differential ranges from $d = -0.282$ (retaining only the two evaluative methods, stance and keywords) to $d = +0.025$ (dropping stance entirely; $p = .15$, a null). The composite is therefore *not* robust to removing the stance method: the aggregate differential is carried by stance and keyword salience, while the two sentiment channels are individually equivalent to zero (Section 4.1). This dependence is expected under the paper’s central finding that the differential is evaluative rather than tonal, but it means the composite should be read as a summary of the evaluative signal diluted by null tonal components, not as evidence that every measurement channel points in the same direction. The composite provides a parsimonious overall test of directional differential treatment, while per-method analyses identify which dimensions drive that aggregate result.

3.4 Media Framing Analysis

Framing is estimated separately from the A–D composite because it measures narrative *type* (which frame is used), not directional evaluative intensity on a single signed scale. For each attributed article, I estimate `frame_*` probabilities and a `dominant_frame` from prosecutor-centered text windows (five-sentence windows around mentions) using BART-MNLI in multi-label mode, with keyword-based fallback scoring when no valid mention window is available or model inference fails. The fallback is not a rare edge case: per the `frame_method` indicator, 6,652 of 12,827

articles (51.9%) receive keyword-based rather than zero-shot frame scores, so the framing results should be read as a hybrid of model-based and dictionary-based measurement; this is a limitation of the framing layer specifically.

The two frame instruments are not on a common scale, so the continuous contrasts are estimated on the zero-shot subset only. The two instruments do not merely differ in accuracy: they report different quantities on incommensurable scales. The zero-shot classifier returns multi-label entailment probabilities, whereas the keyword fallback returns the fraction of dictionary patterns matched, and the resulting score levels differ by roughly a factor of thirty. Mean `frame_human_interest`, for instance, is 0.830 for progressive-attributed and 0.904 for traditional-attributed articles under the classifier, against 0.0256 and 0.0261 under the fallback. Which instrument scores an article is also not random with respect to group: 50.2% of progressive-attributed articles are scored by the zero-shot classifier against 45.9% of traditional-attributed ones, so progressive articles are disproportionately measured on the high-scale instrument. Averaging the two together therefore produces group means that partly reflect instrument assignment rather than content. Accordingly, the continuous `frame_*` contrasts reported in Section 4.6 are estimated on the 6,175 zero-shot-classified articles only, and the pooled probability-score contrasts are reported once, in Section 4.6, as a measurement artifact rather than as a substantive result. The `dominant_frame` contrast is less exposed to the problem, because an `argmax` taken within an article is in-kind across instruments even when the underlying levels are not comparable; but the fallback’s `argmax` is taken over near-zero dictionary match fractions and is correspondingly noisy, so the zero-shot-only chi-square is reported as primary and the pooled chi-square as secondary.

The framing fallback and the attribution fallback are the same articles. This paper reports two fallback-related limitations—the keyword frame fallback just described and the county-and-date attribution fallback described in Section 3.2—and it would be easy to read them as two independent measurement problems with two independent robustness checks. They are not. Crosstabulating the two indicators over the 12,827 analyzed articles gives 6,651 fallback-attributed articles and

6,652 keyword-framed articles with an overlap of 6,637: $Jaccard = 0.996$, $P(\text{keyword frame} \mid \text{fallback attribution}) = 0.998$, and $P(\text{keyword frame} \mid \text{named-mention attribution}) = 0.002$. Only 29 of 12,827 articles fall on one side and not the other. The identity is mechanical once stated: both layers key off the same condition. An article that contains no named prosecutor mention offers the framing classifier no mention window to score, which forces the keyword fallback, and it is the same absence of a named mention that forces attribution to fall back to county-and-date assignment. The practical consequence is reported where the two analyses appear (Section 4.8 and Appendix F): the zero-shot-only framing sample and the exclude-fallback composite sensitivity are two views of one subsample, not independent corroboration.

1. **Accountability:** The prosecutor is held responsible for outcomes (e.g., crime rates, case decisions).
2. **Conflict:** Coverage emphasizes disagreement (prosecutor vs. police, victims, or community).
3. **Consequences:** Focus on policy outcomes and their effects.
4. **Human interest:** Individual stories of victims or defendants as narrative vehicles.
5. **Reform/ideology:** The prosecutor is characterized through an ideological or reformist lens.

Method B and framing both use BART-MNLI but operationalize different constructs: stance asks whether language criticizes or supports a prosecutor, whereas framing asks which narrative lens structures the article. On the model-classified articles, where the frame scores and the stance scores come from the same instrument, pairwise correlations between article-level stance scores and the five frame probabilities run from $r = -0.39$ (accountability) to $r = +0.03$ (human interest), with conflict, consequences, and reform between -0.22 and -0.26 ($n = 6,157$)—far below the $r > 0.8$ threshold for methodological redundancy. The negative values are theoretically expected (critical stance co-occurs with accountability framing) and confirm convergent validity rather than collinearity. Computed on the pooled sample the same correlations run from $r = -0.21$ to $r = -0.10$ ($n = 9,905$), but that version mixes the two frame instruments and is reported only for

completeness.

3.5 Prosecutor-Attributed Theme Detection

Prosecutor-attributed theme detection is estimated separately from both the A–D composite and frame analysis because it targets explicit prosecutor-linked critical narratives rather than general evaluative tone or frame category prevalence. Nine themes are tracked: crime-rising, releasing criminals, victim neglect, police conflict, office dysfunction, soft-on-crime, case dismissal, recall, and public safety failure. To complement Method C’s proximity-based theme signal, the attribution algorithm requires explicit prosecutor-theme linkage via four detection methods: (1) context-aware dictionary patterns requiring both a prosecutor reference and a theme keyword within the same expression (e.g., “crime rising because of the DA” rather than merely “crime” near “prosecutor”), (2) relationship-based regex patterns capturing causal structures linking prosecutors to outcomes, (3) targeted criticism patterns, and (4) sentence-level co-occurrence checks. Method scores (0–25 each) are combined with weights 0.30, 0.35, 0.15, and 0.20, then adjusted by method-agreement rules (single-method caps at 15; $1.1\times$ multiplier for two-method agreement, $1.25\times$ for three or more). The resulting ta_* variables are used in dedicated theme-attribution analyses.

3.6 LLM-Based Structural Content Extraction

The preceding NLP analyses (Methods A–D, framing, and prosecutor-attributed theme detection) quantify evaluative, framing, and attributional signal in text. They are well suited for comparative inference at scale but do not directly enumerate full article structure (who is quoted, what claims are made, and how causal attribution is framed). To capture that layer, I add a complementary LLM structural extraction approach using Google’s **langextract** library (Gemini 2.5 Flash), applied to all attributed articles in the analysis sample ($n = 12,827$, with $n = 6,585$ progressive and $n = 6,242$ traditional).

Langextract is extractive rather than classificatory: outputs are schema-constrained instances linked

to exact source spans. The extraction schema covers source attribution, claims against the prosecutor, causal claims, policy actions, and explicit comparisons. Model behavior is stabilized through fixed prompting and few-shot calibration examples, yielding auditable structural measures that can be aligned with score-based NLP patterns.

Table 2: Complementary measurement layers: NLP methods vs. LLM structural extraction.

Dimension	NLP scoring/detection + framing + theme attribution)	scoring (A–D)	LLM extraction (langextract)	Complementary role
Construct	Evaluative, framing, and attribution signal		Source and claim structure	“How” coverage evaluates vs. “what” coverage contains
Unit	Article/paragraph scores and attribution flags		Span-level extracted instances	Cross-layer triangulation
Output	Continuous indices plus validated indicators		Discrete counts and rates	Mechanism mapping for score differences
Strength	High comparability and statistical power		Source-grounded, auditable outputs	Convergence strengthens interpretation
Limitation	Can obscure specific mechanisms		Depends on schema and calibration choices	Divergence is diagnostically informative

Table 2 sets out how the two layers divide the labor. The NLP methods characterize how coverage evaluates and frames prosecutors, while langextract identifies the structural content through which that coverage is built. When both layers converge, as when the bias-indicator extraction’s article-level score correlates moderately with the main pipeline’s tone–evaluation index ($r = .25$; Appendix B), the evidence is stronger than either method alone. When they diverge, as when the overall bias-indicator contrast is TOST-equivalent within $|d| < 0.2$ (Appendix B) despite significant stance and keyword effects (Section 4.1), the divergence reveals that the differential operates through aggregate framing mechanisms rather than discrete journalistic violations. Main-text structural results appear in Section 4.8; full extraction protocol details remain in Appendices A–B.

3.7 Statistical Approach

I employ six analytical strategies: (1) group-level comparison (Welch’s t -test, Mann–Whitney U , Cohen’s d , Cliff’s delta, bootstrap 95% CIs); (2) same-county paired comparisons (Boudin vs. Jenkins in San Francisco; Price vs. O’Malley in Alameda); (3) OLS regression of bias score on prosecutor ideology, controlling for county, year, and article length, with cluster-robust standard errors by publication; (4) framing differential analysis (chi-square tests on dominant frame frequencies by prosecutor type); (5) segmented interrupted time-series models on monthly county-level outcomes around prosecutor transitions, estimating immediate level shifts (post) and post-transition slope changes (time_after) with HAC-robust inference, complemented by quarterly heterogeneity summaries; and (6) TOST equivalence testing ([Lakens, 2017](#)) with an equivalence bound of $d = 0.2$, applied to the composite and to each A–D component. Group-level bootstrap inference is supplemented by a publication-cluster bootstrap that resamples the 20 outlets with replacement, addressing dependence among articles from the same publication (Section 3.1). Analyses (1), (2), (3), and (5) are reported for the composite and for each A–D component, while (4) is framing-specific. Prosecutor-attributed theme detection uses its own effect-size and prevalence tests, reported in the dedicated theme-attribution subsection.

The $d = 0.2$ bound corresponds to Cohen’s definition of a “small” effect and is standard in equivalence testing applications. Crucially, in a large- N study ($n \approx 13,000$ articles) trivially small effects achieve statistical significance, so the TOST serves as a meaningful practical check: an equivalence result ($p = .003$) indicates the composite effect is not only small by conventional standards but falls within a range generally considered substantively negligible. The per-method analyses are critical: by running each test separately for stance, sentiment, keywords, and document tone, I can identify *which dimensions* of coverage differ by ideology rather than relying on a composite that may average opposing signals to zero. Because the analysis reports many related tests, Benjamini–Hochberg false-discovery-rate adjusted p -values are computed within each test family (per-method comparisons, per-frame contrasts, per-theme prevalence tests, and segmented-ITS

coefficients) and reported in the replication outputs; where the adjustment changes a conclusion, the text says so explicitly. Throughout, p -values below .001 are reported as $p < .001$; exponential notation (e.g., $p < 10^{-123}$) is used selectively where the extreme magnitude is itself informative.

4 Results

4.1 Bias Is Not One Thing: Per-Method Decomposition

The central finding of this study is that “media bias” toward prosecutors is not a single phenomenon. The four bias detection methods, each measuring a distinct dimension of coverage, produce sharply divergent results (Figure 1):

- **Stance classification (B)** produces the largest of the four method effects ($d = -0.337$, $p < .001$): articles about progressive prosecutors contain significantly more critical and fewer defensive paragraphs. This method captures evaluative framing: whether coverage *judges* the prosecutor favorably or unfavorably.
- **Keyword analysis (C)** shows a small-to-moderate effect ($d = -0.203$, $p < .001$): anti-prosecutor themes (soft-on-crime, crime-rising, recall) appear more frequently near mentions of progressive prosecutors. This method captures thematic emphasis.
- **Aspect-based sentiment (A)** shows a tiny *reversed* effect ($d = 0.047$, $p = .018$): the emotional tone surrounding mentions of progressive prosecutors is, if anything, marginally more positive, and the difference is affirmatively bounded within the equivalence range (below). This method captures emotional valence.
- **Document-level sentiment (D)** shows a small reversed effect ($d = 0.049$, $p = .005$): articles about traditional prosecutors are slightly *more* negative in overall tone, reflecting the inherent negativity of crime reporting rather than prosecutor-specific bias.

The tone–evaluation split is formally symmetric under per-method TOST equivalence testing at the $|d| = 0.2$ bound: aspect-based sentiment and document-level sentiment are each statistically

equivalent to zero ($p_{\text{TOST}} < .001$ for both; although each tonal contrast is nominally significant in this large sample, both are bounded within the negligible range), whereas stance and keyword salience fail the same equivalence test ($p_{\text{TOST}} = 1.0$ and $.56$, respectively). Tonal equivalence is thus an affirmative statistical finding, not merely an absence of significance.

Method C is heavily zero-inflated, and the zeros work against the reported effect. One objection to the keyword result deserves a pre-emptive answer, because the mechanism is real. Method C scores anti-prosecutor theme density within windows around named prosecutor mentions, so an article with no named mention has no window and receives exactly 0.0—the most favorable value on the measure’s $[-1, 0]$ range. This is a structural zero, not an observation of neutral coverage, and it is common: 99.97% of the 6,651 fallback-attributed articles score exactly zero, and the structural-zero rate differs by group in the full sample (84.9% progressive vs. 91.4% traditional). A reader could reasonably suspect that the group difference in how often the instrument is silent is manufacturing the group difference in its readings. It is not, and the direction is the opposite of the concern. Decomposing the full-sample differential ($d = -0.203$, $p = 7.9 \times 10^{-31}$; $n = 6,585/6,242$) by whether a named mention was available gives $d = -0.264$ ($p = 8.3 \times 10^{-26}$; $n = 3,322/2,854$) among articles the instrument can actually measure and a null among the all-zero fallback articles ($d = +0.034$, $p = .16$). Even within the measurable subsample the score remains zero-inflated and unequally so (70.0% progressive vs. 81.3% traditional), yet the differential is larger there than in the pooled figure. The structural zeros dilute the keyword differential toward zero rather than creating it; the reported $d = -0.203$ is an attenuated version of the $d = -0.264$ that holds where the measurement is defined.

The divergence between stance ($d = -0.337$) and sentiment ($d = 0.047$) is the study’s key conceptual finding: media coverage of progressive prosecutors maintains a neutral emotional tone while adopting a critical evaluative stance. A diagnostic visualization of this tone–evaluation split—a KDE contour plot of aspect sentiment against stance—is reported in Figure 13 (Appendix C).

This pattern has a direct methodological implication: studies relying solely on sentiment analysis

would conclude that media treatment of prosecutors is equivalent. The differential emerges only when measurement captures evaluative framing, a dimension invisible to emotional tone detection.

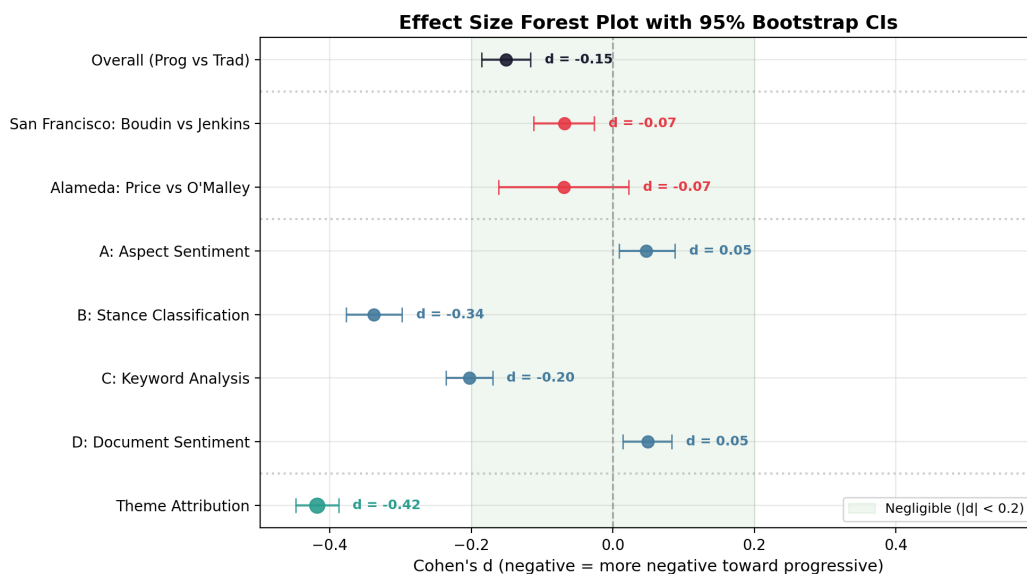


Figure 1: Effect size forest plot with 95% bootstrap confidence intervals. The green band marks the negligible effect zone ($|d| < 0.2$). Results are grouped by analysis type: overall, paired-county, per-method, and theme attribution. Theme attribution ($d = -0.418$) is the largest effect in magnitude plotted here, but it is substantially event-driven and reverses sign once recall- and crime-rising-flagged articles are removed, so the plotted magnitude is an upper bound (Section 4.7).

4.2 Same-County Paired Comparisons by Method

The within-county design provides the strongest quasi-experimental identification. Per-method analysis reveals *which dimensions* of coverage differ within each jurisdiction (Table 3):

San Francisco (Boudin vs. Jenkins). Stance classification detects significant differential treatment ($d = -0.197$, $p < .001$) and keyword analysis confirms it ($d = -0.093$, $p < .001$). Aspect sentiment ($d = 0.058$, $p = .020$) and document sentiment ($d = 0.069$, $p = .002$) show small *reversed* (positive) differences. The bias is evaluative, not tonal.

Alameda County (Price vs. O'Malley). On a composite measure, this pair shows a non-significant difference ($d = -0.069$, $p = .138$), an apparent null. However, decomposition by method reveals

that the null is an artifact of aggregation: stance classification ($d = -0.272$, $p < .001$) and keyword analysis ($d = -0.381$, $p < .001$) both detect substantial differential treatment, while sentiment ($d = 0.045$, $p = .381$) and document tone ($d = 0.105$, $p = .017$) pull in the *opposite* direction. The composite averages these opposing signals into a near-zero result, masking real differential treatment that operates through evaluative framing.

Table 3: Per-method Cohen’s d for same-county paired comparisons. Significant results ($p < .05$) in bold.

Method	San Francisco		Alameda	
	d	p	d	p
B: Stance	−0.197	< .001	−0.272	< .001
C: Keywords	−0.093	< .001	−0.381	< .001
A: Aspect sentiment	0.058	.020	0.045	.381
D: Document sentiment	0.069	.002	0.105	.017
Composite	−0.068	.002	−0.069	.138

Figure 2 displays per-prosecutor mean composite scores with 95% confidence intervals. Figure 3 presents the paired county comparisons.

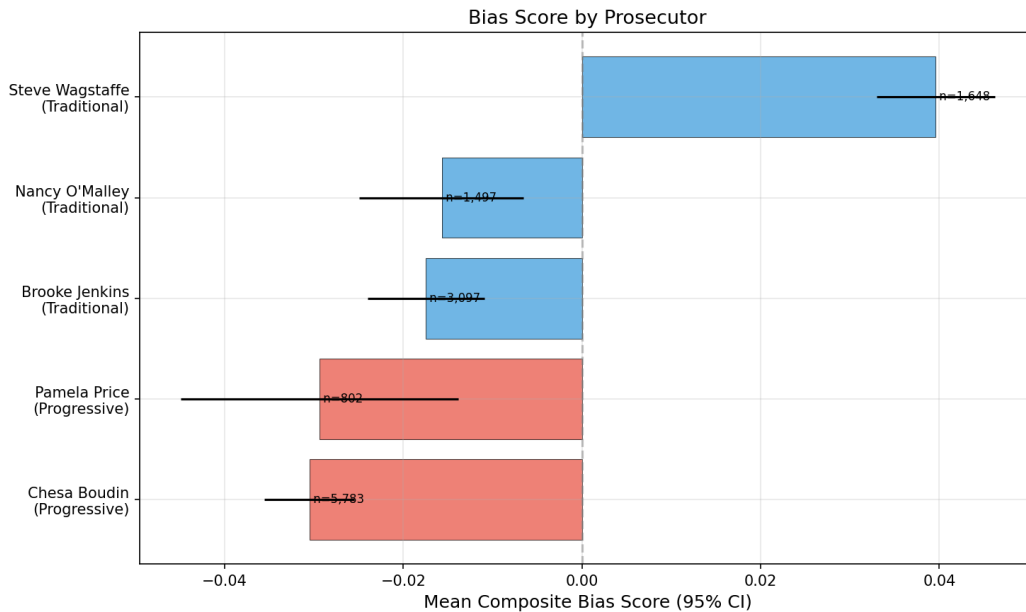


Figure 2: Mean aggregate tone-evaluation index by prosecutor with 95% confidence intervals. Red bars indicate progressive prosecutors; blue bars indicate traditional prosecutors.

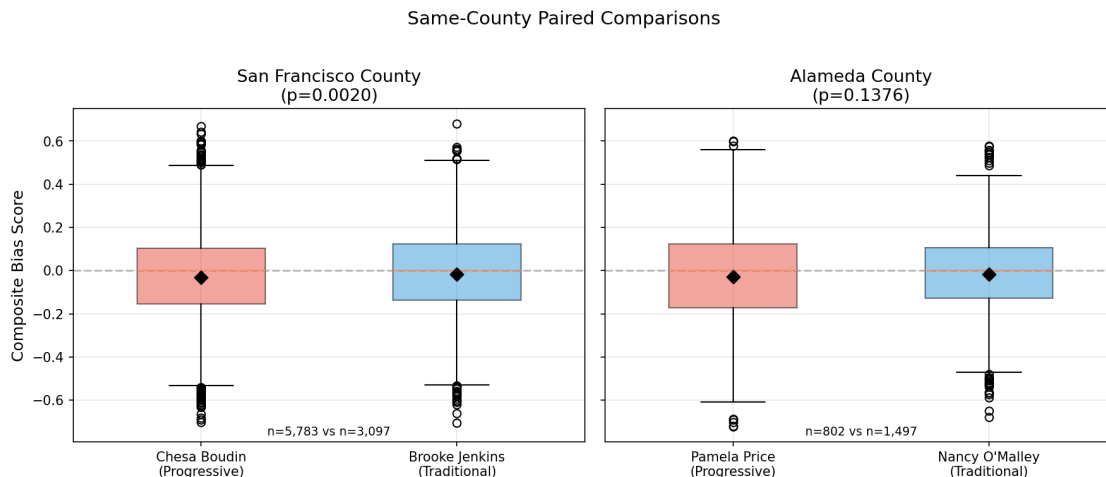


Figure 3: Same-county paired comparisons: San Francisco (Boudin vs. Jenkins) and Alameda (Price vs. O'Malley). Diamond markers indicate group means.

4.3 Regression Analysis by Method

To test whether each bias dimension survives controls for county, year, article length, and publication clustering, I estimate separate OLS regressions for each method (Figure 4; full coefficient table in Table 9, Appendix C).

Stance classification and keyword analysis, the evaluative and thematic dimensions, both show significant progressive coefficients that survive all controls. Aspect sentiment and document sentiment do not. The composite regression falls to non-significance ($p = .104$) precisely because it averages the strong evaluative signal with the null tonal signal. Running regressions per method resolves this: evaluative bias is robust to controls; tonal bias does not exist.

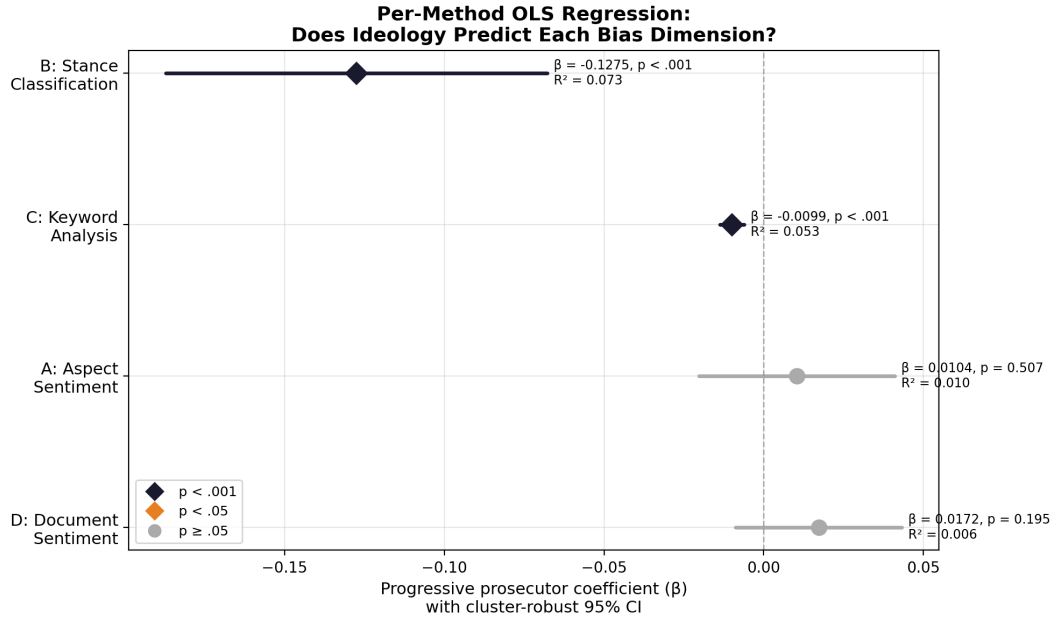


Figure 4: Per-method OLS regression: progressive prosecutor coefficient with 95% cluster-robust CIs. Stance and keyword methods survive controls; sentiment methods do not.

4.4 Temporal Dynamics by Method

Per-method temporal analysis combines segmented interrupted time-series modeling with quarterly decomposition to test whether the evaluative bias identified above is stable or event-driven (Figure 5).

Segmented interrupted time series. Monthly segmented models estimate transition-date level changes and post-transition slope changes for each outcome (Appendix G, Table 14, Figure 16). In San Francisco, the composite shows a modest positive level shift that does not reach significance ($p = .070$) and a nominally significant positive post-transition slope, consistent with movement toward less negative coverage after the Boudin-to-Jenkins transition. In Alameda, the composite and stance outcomes show immediate negative level shifts at the O’Malley-to-Price transition with little compensating slope change, consistent with a step increase in critical evaluative coverage. These composite-level ITS coefficients should, however, be read as suggestive: after Benjamini–Hochberg correction across the full 30-test ITS family, the San Francisco composite slope change

and the Alameda composite level and 12-month horizon effects are no longer significant, whereas the stance and keyword effects in both counties, and the San Francisco composite 12-month horizon effect, largely survive correction (Appendix G). A controlled specification that differences each treated county against San Mateo County (Steve Wagstaffe), which experienced no prosecutor transition, confirms the Alameda level shift ($p = .003$) but not the San Francisco one ($p = .26$; Appendix G). Across transitions, evaluative measures (especially stance, then keywords) carry the temporal signal, whereas sentiment measures remain weak and inconsistent.

Quarterly heterogeneity. Figure 5 decomposes the temporal patterns of each method across 21 quarters. Stance classification ($SD = 0.32$, range $[-0.66, +0.30]$) and keyword analysis ($SD = 0.36$, range $[-0.86, +0.54]$) show dramatic quarterly variation. Both series fluctuate in sign through 2019–2020, with quarters as deep as any that follow (stance = -0.27 in 2019Q4 and $hBc0.30$ in 2020Q1; keywords $hBc0.46$ in 2019Q1 and $hBc0.47$ in 2019Q4), before settling into sustained negative values from 2021Q4 onward. The early quarters are volatile rather than neutral, and the shift is one of persistence rather than of level. The deepest stance quarters span the Boudin recall and its immediate aftermath (2021Q4–2022Q3, $d = -0.46$ to -0.65) and then again 2023Q4–2024Q1 (-0.63 and -0.66); the deepest keyword quarters fall just after the June 2022 recall (2022Q3–Q4, $d = -0.73$ and -0.86). Neither series peaks during the Price recall quarters themselves (2023Q2–Q3), where the keyword differential is among its shallowest post-2021 values (-0.12 and -0.16). Aspect sentiment ($SD = 0.28$, range $[-0.32, +0.77]$) and document sentiment ($SD = 0.22$, range $[-0.15, +0.69]$) vary comparably in magnitude—aspect sentiment’s quarterly range is in fact wider than stance’s—but the variation is not tied to political events: both sentiment series take their largest values in the four pre-tenure 2019 quarters and run much closer to zero from 2020 onward, whereas the evaluative series deepen through the recall periods. The monthly time series (Figure 6) overlays a coverage-volume-weighted three-month rolling average alongside the raw monthly means; the weighted and unweighted trends are nearly identical, confirming that the temporal patterns are not artifacts of high-variance low-coverage months.

This per-method temporal decomposition resolves a key ambiguity in the aggregate analysis. The composite quarterly d (range $[-0.38, +0.86]$, $SD = 0.39$) appears to show dramatic cyclical variation, but this aggregate masks the distinct temporal signatures of its component methods. Evaluative bias (stance, keywords) is event-driven, surging during recall campaigns and subsiding between them. Tonal bias (sentiment) is absent throughout. The aggregate temporal variation is driven entirely by the evaluative dimensions.

Per-Method Temporal Heterogeneity: Quarterly Effect Sizes

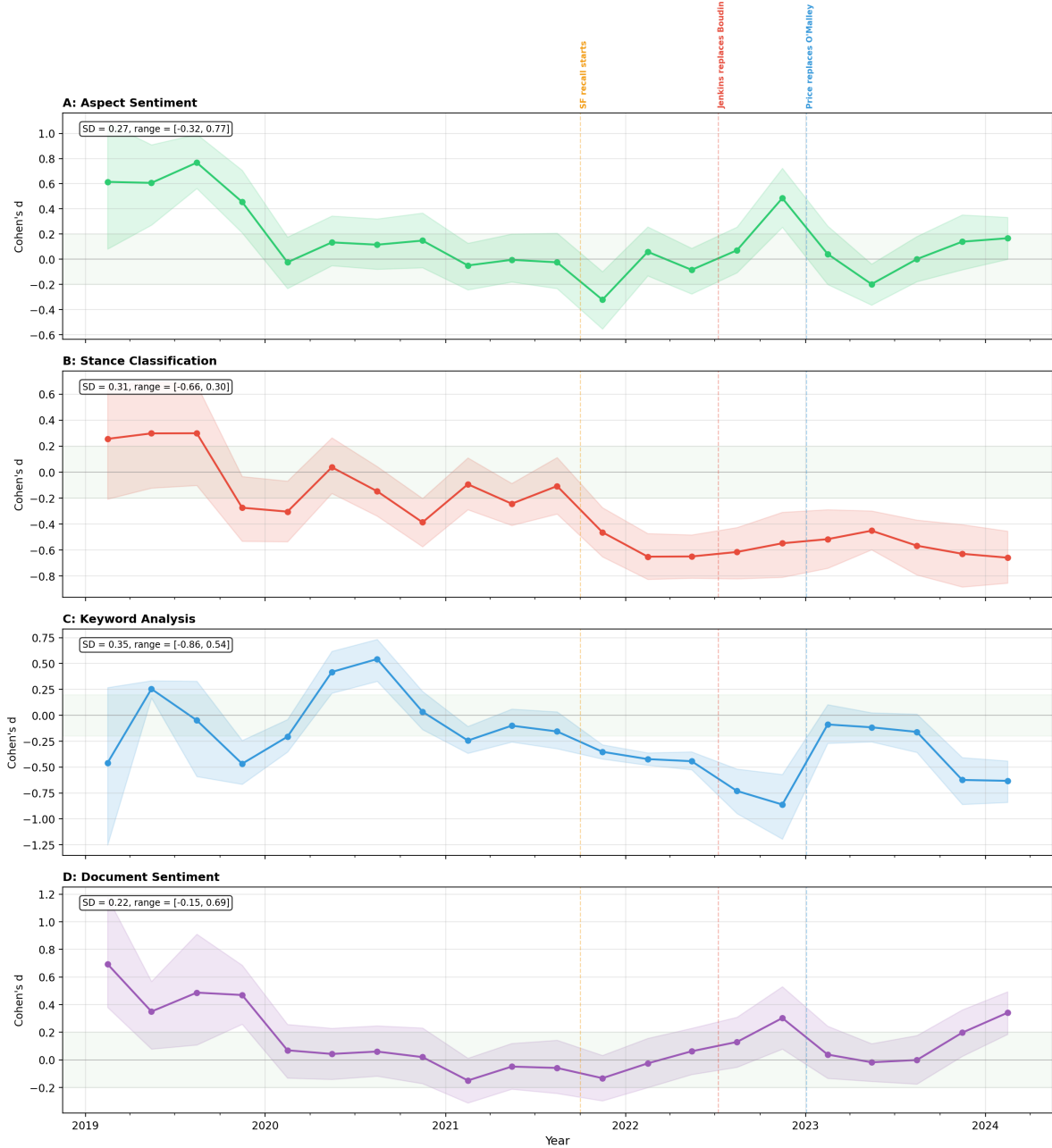


Figure 5: Per-method quarterly Cohen's d (progressive vs. traditional) with 95% bootstrap CIs. The green band marks the negligible zone ($|d| < 0.2$). Dashed lines mark prosecutor transitions. Stance and keyword methods show event-driven variation; sentiment methods show no event-linked pattern.

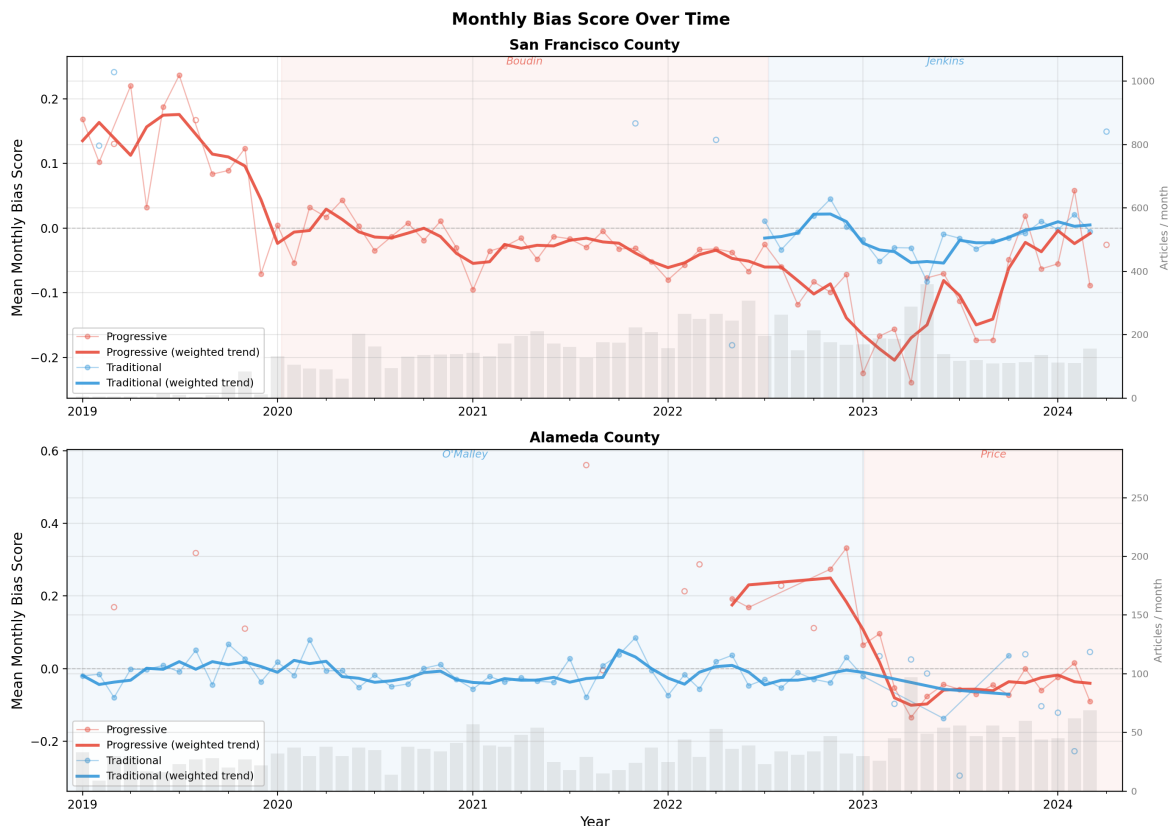


Figure 6: Monthly mean composite scores for San Francisco (top) and Alameda County (bottom), 2019–2024. Faint dots and lines show raw monthly means; thick lines show coverage-volume-weighted three-month rolling averages. Open circles mark months with fewer than five articles. Shaded bands indicate prosecutor tenures; red = progressive, blue = traditional.

4.5 Composite Summary

When the four methods are combined into a weighted composite (weights: 0.35 sentiment, 0.30 stance, 0.20 keywords, 0.15 document), the overall group comparison yields $d = -0.151$ ($p < .001$; 95% bootstrap CI for the raw mean difference $[-0.035, -0.022]$), with a TOST equivalence test confirming the effect falls within conventional equivalence bounds ($p = .003$ at $d = 0.2$; see Section 3.7 for bound justification). Because articles cluster within outlets, I also re-estimate the group contrast with a publication-cluster bootstrap that resamples the 20 outlets with replacement: the 95% CI for the raw mean difference is $[-0.0475, -0.0074]$, with a two-sided bootstrap $p \approx .011$. The contrast therefore survives outlet-level dependence, though the evidence is far less

extreme than the article-level i.i.d. p -value suggests. The composite is also insensitive to quoted speech: a sensitivity variant that masks keyword matches inside quotations (Method C’s channel) yields an essentially identical estimate ($d = -0.151$, $p < .001$). The composite is negative and significant under every alternative weighting scheme that retains the stance method, and null when stance is dropped (Section 3.3). This composite serves as a proxy for the aggregate reader experience: a reader does not consume “stance” and “sentiment” as separate channels but is exposed to both the emotional tone and the evaluative framing of an article simultaneously. The composite approximates this net impression and provides a single dependent variable for the time series and overall comparison (Figure 7). However, the per-method results above demonstrate that this aggregate figure obscures the distinct mechanisms at play: it averages a large evaluative effect with a null tonal effect, producing a small overall number that understates the stance differential and overstates the sentiment contribution.

The pooled composite is inflated by county composition. A leave-one-prosecutor-out analysis (Appendix E, Table 11) shows that the pooled estimate is not equally supported by all five prosecutors. Dropping Boudin, Price, or O’Malley leaves it within 15% of the full-sample value, and dropping Jenkins raises it to $d = -0.233$; but dropping Steve Wagstaffe cuts it to $d = -0.070$, 46% of the full-sample effect. The reason is that Wagstaffe’s coverage is markedly more favorable than any other prosecutor’s on both the composite and stance; the per-prosecutor means are reported there. He is not a representative traditional prosecutor: he is the only prosecutor in the study from San Mateo County, the only county with no transition, and his 1,648 articles supply 26% of the traditional baseline. Including him widens the pooled gap.

Three routes to the same quantity that do not compare across counties agree closely. Dropping Wagstaffe from the pooled contrast gives $d = -0.070$; the San Francisco within-county pair gives $d = -0.068$ ($p = .0020$); the Alameda within-county pair gives $d = -0.069$ ($p = .138$). The pooled cross-sectional figure of $d = -0.151$ is roughly double all three. Since the within-county design is this study’s strongest identification (Section 4.2), the honest reading is that $d \approx -0.07$

is the composite differential the credible designs converge on, and the pooled figure is an upper bound whose excess is attributable to cross-county heterogeneity in coverage levels rather than to prosecutor ideology. This does not overturn the paper’s per-method conclusion—the within-county stance and keyword effects are larger than the within-county composite and survive controls—but it does mean the pooled composite headline should not be read as the effect magnitude.

The remaining robustness checks divide cleanly, and not all of them favor the reported estimates. The relevance- and period-based restrictions preserve or increase effect magnitudes, indicating attenuation rather than artifact in the full-sample estimates, though they are nested rather than independent restrictions (Appendix F). One pair is nearly identical: the exclude-fallback sample and the model-classified subset that carries the primary framing analysis are one subsample viewed through two instruments (Section 3.4), so the framing result and the composite fallback check count as one piece of evidence, not two.

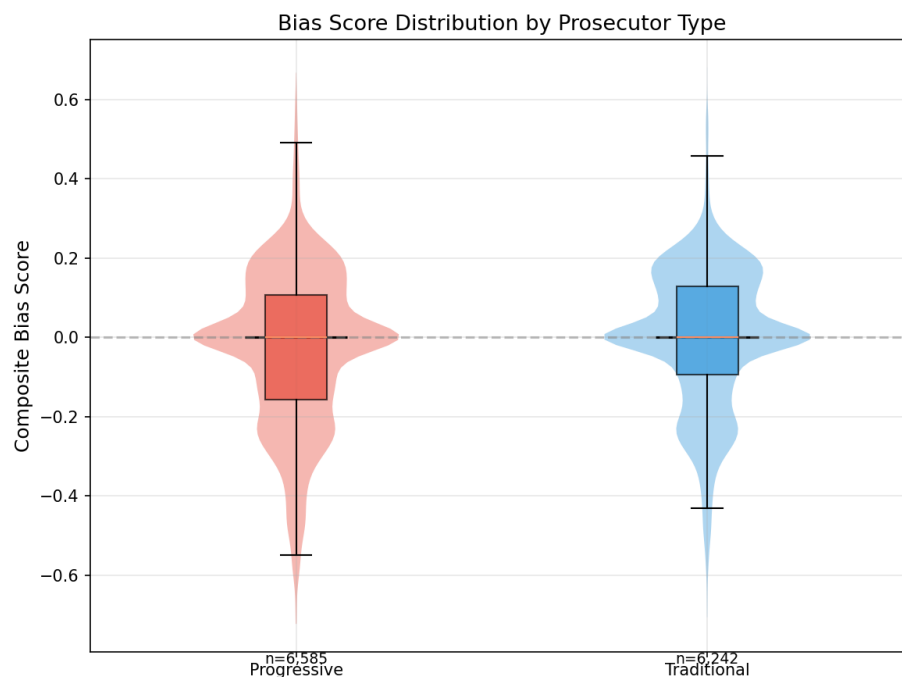


Figure 7: Aggregate tone-evaluation index distributions by prosecutor ideology. Violin plots show the full distribution; inner box plots show the interquartile range. Negative scores indicate more negative coverage.

4.6 Framing Analysis

The distribution of dominant media frames differs significantly between progressive and traditional prosecutors. The primary estimate is taken on the 6,175 articles scored by the zero-shot classifier, for the scale reasons set out in Section 3.4: $\chi^2(4) = 492.10$, $p < 10^{-104}$, Cramér’s $V = 0.282$. Pooling the keyword-fallback articles back in gives a same-signed but weaker contrast, $\chi^2(4) = 395.66$, $p < 10^{-83}$, $V = 0.214$. The pooled chi-square is reported as secondary rather than discarded, because argmax within an article is at least in-kind across the two instruments; but the fallback’s dominant frame is an argmax over near-zero dictionary match fractions, so roughly half of the pooled table’s rows are assigned on a far noisier basis than the other half. Table 4 reports both sets of rates (see also Figure 8).

Table 4: Dominant media frame distribution by prosecutor ideology. The model-classified columns ($n = 6,175$) are the primary estimate; the pooled columns ($n = 12,827$) add the keyword-fallback articles and are reported as secondary.

Frame	Model-classified (primary)		Pooled (secondary)	
	Progressive	Traditional	Progressive	Traditional
Accountability	40.4%	18.7%	41.1%	24.6%
Conflict	37.0%	47.7%	29.8%	36.6%
Human interest	17.4%	32.3%	21.4%	33.6%
Reform	5.0%	1.2%	4.2%	1.2%
Consequences	0.2%	0.1%	3.5%	3.9%

On the primary model-classified sample, progressive prosecutors are more than twice as likely to be covered through an accountability frame (40.4% vs. 18.7% of articles; Cohen’s $h = 0.48$) and more than four times as likely through a reform frame (5.0% vs. 1.2%, $h = 0.23$). Conversely, traditional prosecutors receive substantially more human-interest framing as the dominant frame (32.3% vs. 17.4%, $h = -0.35$): stories organized around individual victims, defendants, or community members rather than evaluations of prosecutorial performance. Traditional prosecutors are also more likely to receive conflict as the dominant frame (47.7% vs. 37.0%, $h = -0.22$), while dominant consequences framing is vanishingly rare in this subset and essentially equal across groups (0.2% vs. 0.1%, $h = 0.02$). The continuous frame-probability contrasts, estimated on the

same model-classified articles, align in sign with the assignment rates at every frame: progressive coverage scores higher on reform ($d = 0.607$, $p < 10^{-119}$), accountability ($d = 0.249$, $p < 10^{-21}$), and consequences ($d = 0.216$, $p < 10^{-16}$), while traditional coverage scores higher on human interest ($d = -0.413$, $p < 10^{-59}$) and conflict ($d = -0.119$, $p < .001$). All five remain significant after Benjamini–Hochberg correction across the five-frame family. Rate effect sizes are Cohen’s h on per-frame dominant-frame assignment rates, matching the model-classified proportions in Table 4; the d values refer to the underlying frame probability scores.

The pooled frame-probability contrasts are an instrument-mixing artifact and are withdrawn as substantive results. An earlier version of this paper reported the continuous frame contrasts on the pooled sample—accountability $d = 0.133$, reform $d = 0.269$, consequences $d = 0.174$, conflict $d = 0.074$, human interest $d = 0.001$ —and explained the gap between those values and the model-classified ones as attenuation by the keyword fallback. That explanation was wrong, and the diagnostic that shows it is worth reporting rather than quietly dropping (`output/06_stats_results.json`, `framing.instrument_mixing_diagnostic`). Table 5 decomposes each pooled Cohen’s d into the two within-instrument values it supposedly combines. For accountability, consequences, and reform the pooled value falls between the two subgroup values, as a weighted combination should. For conflict and human interest it does not: the pooled conflict d is $+0.074$ while its two parts are -0.119 (zero-shot) and $+0.035$ (keyword fallback), and the pooled human-interest d is $+0.001$ while its parts are -0.413 and -0.008 . A genuine weighted combination of two subgroups measured on a common scale cannot lie outside the range those subgroups span. That two of the five do is direct evidence that the pooled figures are not measurements of one quantity at all. It also rules out the attenuation story specifically: attenuation can only pull an estimate toward zero, never push it past both of its parts.

The mechanism is the one set out in Section 3.4: the two instruments differ in level by roughly thirtyfold, and instrument assignment is correlated with prosecutor type, so pooled means partly track which instrument scored an article. The pooled probability contrasts are reported here once,

Table 5: Per-frame Cohen’s d (progressive minus traditional), pooled versus within-instrument. For conflict and human interest the pooled value lies *outside* the range spanned by the two subgroups (marked †), which a weighted combination on a common scale cannot do. Source: `output/06_stats_results.json, framing.instrument_mixing_diagnostic`.

Frame	Pooled d	Zero-shot d	Keyword-fallback d
Accountability	+0.133	+0.249	+0.074
Conflict	+0.074 †	−0.119	+0.035
Consequences	+0.174	+0.216	+0.046
Human interest	+0.001 †	−0.413	−0.008
Reform	+0.269	+0.607	+0.045

as the artifact they are, and carry no substantive claim anywhere in the paper. The honest summary is not that the pooled figures are conservative but that they are uninterpretable, and that the model-classified figures—which are the *larger* ones on four of the five frames—are the ones to read.

Two cautions attach to the model-classified sample even in its primary role. First, this subsample is, up to 29 articles, the same one used for the exclude-fallback composite sensitivity (Section 3.4; Appendix F), so the framing result and that sensitivity do not independently corroborate one another. Second, and for the same reason, this subset is not a random half of the corpus but the half in which prosecutors are named, so part of the reason its differentials exceed the pooled ones reflects higher prosecutor centrality and not only cleaner frame measurement.

These framing patterns are consistent with the stance classification finding: the media evaluates progressive prosecutors through accountability and ideological lenses while presenting traditional prosecutors through less evaluative, story-driven narratives.

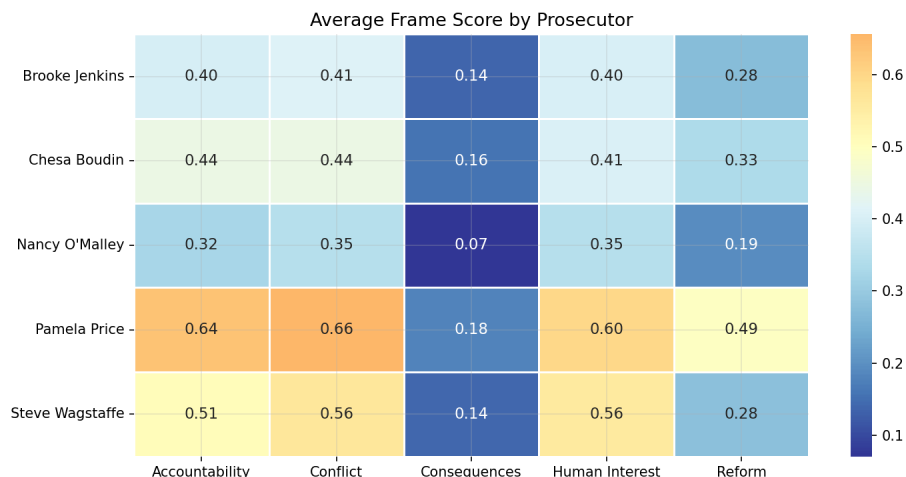


Figure 8: Heatmap of average frame scores by prosecutor. Warmer colors indicate higher frame prevalence. Progressive prosecutors (Boudin, Price) score higher on accountability and reform; traditional prosecutors (Jenkins, O’Malley, Wagstaffe) score higher on human interest. Two cautions. Values are average frame *scores*, whereas Table 4 reports dominant frame assignments (the frame with the highest score per article). And the cell means pool both frame instruments, which are on incommensurable scales (Table 5); per-prosecutor differences in the share of articles reaching the zero-shot classifier therefore contribute to the contrasts shown here, so the figure should be read as illustrative of frame ordering within a prosecutor rather than as a between-prosecutor effect estimate.

4.7 Theme Prevalence and Prosecutor-Attributed Themes

This subsection reports two complementary theme analyses. *Part A (Method C descriptive prevalence)*: Figure 9 presents keyword prevalence from the Method C dictionaries, measured as local term density within three-sentence windows around prosecutor mentions. By this measure, recall-related language dominates progressive prosecutor coverage (9.5% vs. 2.7% for traditional), driven overwhelmingly by Boudin’s 2022 recall campaign. After recall, the most prevalent anti-prosecutor themes for progressive coverage are soft-on-crime (1.6% vs. 0.7%), public-safety-failure (1.2% vs. 0.3%), and crime-rising (0.9% vs. 0.2%). The “releasing criminals” theme is the only category where traditional prosecutors receive higher prevalence (5.5% vs. 4.2%), likely reflecting routine bail and parole coverage rather than ideologically targeted framing.

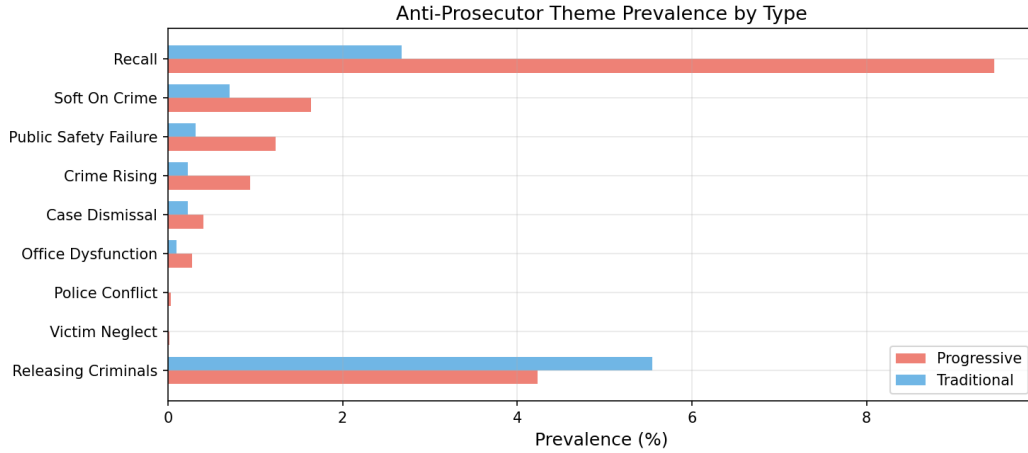


Figure 9: Anti-prosecutor theme prevalence by ideology. Themes are sorted by the difference between progressive and traditional prevalence rates.

Part B (prosecutor-attributed theme model): The multi-method prosecutor-attributed theme analysis requires themes to be explicitly linked to prosecutors through compound regex patterns rather than simple keyword proximity. This stricter attribution measure provides the strongest evidence of differential thematic coverage. Progressive prosecutors receive significantly higher theme attribution scores ($M = 1.68$, $SD = 2.99$) than traditional prosecutors ($M = 0.67$, $SD = 1.65$), $t(10,359) = 24.01$, $p < 10^{-123}$, $d = -0.418$, 95% bootstrap CI for the mean difference $[0.94, 1.10]$. This is the largest raw effect in magnitude among the study’s pooled group contrasts, exceeding the stance classification effect ($|d| = 0.337$); the largest raw effect anywhere in the paper is the Alameda within-county theme contrast reported below ($|d| = 0.68$, reported there in the progressive-minus-traditional direction), which is the same measure applied within a single jurisdiction. It is not, however, the study’s largest *differential-treatment* effect: the decomposition reported below shows that most of it is carried by two themes that track events which actually differed between the two groups, and that the remainder is small and slightly reversed. The differential is not an artifact of quoted speech: excluding theme matches inside quotations, which restricts the comparison to the outlet’s own voice, leaves the effect essentially unchanged ($d = -0.40$, $p < 10^{-115}$).

Per-theme analysis (Figure 10) uses a different metric: the percentage of articles in which *any* of

the four attribution methods flags the theme, producing higher prevalence rates than the keyword-proximity measure above. By this article-level flag, recall-related themes appear 4.47 times more often in progressive prosecutor coverage (18.3% vs. 4.1%, $\chi^2 = 639.6$, $p < 10^{-140}$, Cramér's $V = 0.22$). Crime-rising themes appear 1.80 times more often (20.7% vs. 11.5%, $\chi^2 = 200.1$, $p < 10^{-44}$), and soft-on-crime framing is 3.84 times more prevalent ($\chi^2 = 62.2$, $p < 10^{-14}$). Seven of the nine themes differ significantly after Benjamini–Hochberg correction across the theme family; the exceptions are releasing-criminals and the near-empty office-dysfunction category. The paired county theme comparisons confirm: in San Francisco, Boudin carries a higher attributed-theme load than Jenkins ($d = -0.27$, $p < .001$), and in Alameda, Price carries a higher load than O'Malley ($d = -0.68$, $p < .001$), the study's largest within-county effect in magnitude. The corresponding breakdown by individual prosecutor rather than by ideology group is shown in Figure 14 (Appendix C).

Event decomposition of the theme differential. The two themes with the largest between-group prevalence gaps—recall and crime-rising—are also the two least interpretable as differential treatment, because they track events that genuinely differed between the groups. Both progressive prosecutors in the study faced actual recall campaigns—the June 2022 Boudin recall and the campaign against Price that began in 2023—while no traditional prosecutor did. Covering a real recall campaign is not evidence that comparable officials are treated differently, and the same logic applies more weakly to crime-rising themes, which track a genuine public debate over crime trends during these tenures. I therefore decompose the attributed-theme differential by leave-one-theme-out and by date window. All figures below follow the paper's sign convention, in which a negative d indicates a higher attributed-theme load for progressive prosecutors, and each exclusion drops the articles flagged with the named theme rather than merely zeroing that theme's contribution. Against the full-sample $d = -0.418$ ($p = 5.15 \times 10^{-124}$; $n = 6,585$ progressive, 6,242 traditional), excluding articles flagged with the recall theme (11.4% of the sample) leaves $d = -0.096$ ($p = 5.24 \times 10^{-7}$; $n = 5,379/5,986$), and excluding articles flagged with crime-rising (16.2%) leaves $d = -0.300$ ($p = 3.59 \times 10^{-52}$; $n = 5,221/5,525$). Excluding articles

flagged with *either* theme (22.9% of the sample) reverses the sign of the differential: $d = +0.088$ ($p = 9.24 \times 10^{-6}$; $n = 4,519/5,374$), meaning that in the remaining coverage traditional prosecutors carry a marginally *higher* attributed-theme load than progressive prosecutors. The reversal is specific to these two themes. Dropping any one of the other seven leaves the effect essentially intact, from $d = -0.396$ (public-safety-failure, 1.6% of articles dropped) to $d = -0.456$ (case-dismissal, 4.9% dropped), all within 0.04 of the full-sample value. A date-based version of the same test gives a consistent answer: excluding the two recall-campaign windows entirely (2021-06-01 to 2022-06-30 in San Francisco and 2023-08-01 to 2024-12-31 in Alameda, together 39.1% of the sample) leaves $d = -0.197$ ($p = 1.98 \times 10^{-16}$; $n = 3,311/4,503$)—about half the full-sample magnitude, and larger than the theme-based exclusion because the date windows also remove non-recall coverage from those months. The reading this supports is the one the temporal analysis already implies: a large share of what the theme layer measures is responsiveness to events that differed between the groups, not differential framing of like-for-like situations. The within-county theme contrasts inherit the same caveat, most sharply in Alameda, where Price’s tenure overlaps the recall campaign against her.

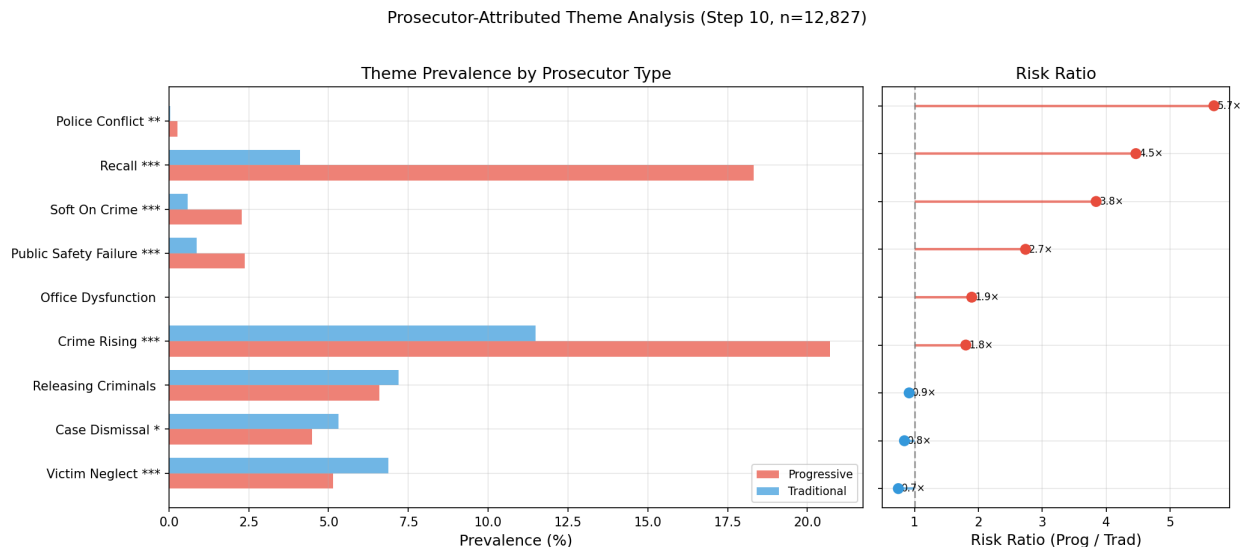


Figure 10: Prosecutor-attributed theme prevalence by ideology (left) and risk ratios (right). Significance: *** $p < .001$, ** $p < .01$, * $p < .05$. Themes sorted by risk ratio.

Multi-method validation shows that 38.2% of articles with any detected theme trigger two or more independent detection methods, confirming convergent evidence rather than single-method artifacts (Figure 15, Appendix C).

4.8 Structural Source Ecology and Claims (Full-Corpus LLM Extraction)

After establishing evaluative differentials with NLP scoring, this section identifies the structural content patterns that co-occur with those differentials. Source types differ strongly by prosecutor ideology ($\chi^2(9) = 1802.1$, $p \approx 0$; Figure 11), with progressive prosecutor coverage drawing disproportionately on advocacy voices ($2.4\times$), journalist commentary ($1.5\times$), politicians ($1.5\times$), and experts ($1.4\times$) relative to traditional coverage. Human validation of extraction recall (Section 4.9) finds that unnamed and documentary attribution is disproportionately missed but that recall failure is symmetric across prosecutor types; the between-group ratios reported here are therefore the robust quantity, while the underlying composition understates documentary sources (see Appendix A).

Table 6: Source stance per article by prosecutor ideology (full extraction, $n = 12,827$). The d column is a pooled-SD Cohen’s d on per-article counts of stance-labeled source attributions, computed from `output/08_extractions_summary.parquet` over the 12,821 articles with successful extraction runs ($n = 6,585$ progressive, 6,236 traditional); positive d indicates a higher per-article count in progressive coverage. All three Welch tests are significant at $p < 10^{-15}$, which at this sample size is uninformative—the effect sizes are the quantity of interest, and all three fall inside the $|d| < 0.2$ negligible band or just outside it.

Source Stance	Prog./art.	Trad./art.	d	Ratio
Critical	1.71	1.28	+0.15	$1.3\times$
Supportive	1.42	0.95	+0.22	$1.5\times$
Neutral	3.17	3.88	−0.21	$0.8\times$

Progressive prosecutors attract both more critical and more supportive source attributions per article, while traditional prosecutors receive more neutral sourcing. The largest absolute source-rate difference remains prosecutor self-quotation (2.09 per article for progressives vs. 2.59 for traditional), indicating that traditional coverage is more often anchored in the officeholder’s own framing while progressive coverage draws more external contestation.

Causal attributions show the same directional asymmetry: “prosecutor caused harm” appears at 0.42 per progressive article versus 0.24 per traditional article (1.7 $\times$), while “prosecutor helped” also appears slightly more often for progressives (0.15 vs. 0.13, 1.2 $\times$). For claim content, policy and performance criticism show the largest differentials (policy: 0.49 vs. 0.31 per article, 1.6 $\times$; performance: 0.45 vs. 0.31, 1.5 $\times$), while character and competence claims are comparatively symmetric (1.2 $\times$ and 1.1 $\times$).

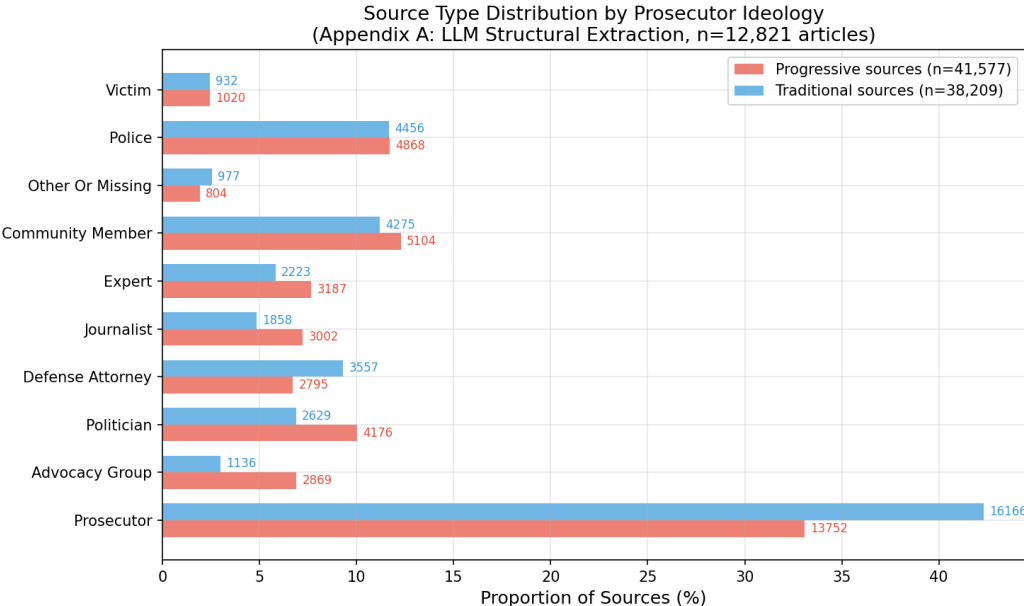


Figure 11: Source type distribution by prosecutor ideology from LLM structural extraction ($n = 12,827$ articles). Bars show proportions of total sources within each group; annotations show raw counts. The largest relative differentials are in advocacy groups (2.4 $\times$), journalist commentary (1.5 $\times$), politicians (1.5 $\times$), and expert sources (1.4 $\times$).

These structural patterns are consistent with the evaluative-framing findings but remain descriptive rather than causal.

4.9 Human Validation of Automated Measures

A trained research assistant, working independently of the author but not blind to the model, completed a four-packet validation program: a 100-article stance-and-frame sample, a 160-extraction audit spanning 142 articles, a 120-article case-type coding task, and a 40-article extraction-recall

audit. These 402 article-slots cover 373 distinct articles, because 23 articles were coded in two of the packets and 3 in all three; the first three packets alone cover 333 distinct articles, which is the denominator for the pooled relevance benchmark reported below. The findings are reported here by the measure each bears on, in the order those measures appear in this section, rather than by packet. Non-blinding is a design caveat that applies throughout: every packet displayed the model’s labels alongside the article text, so the coder formed her judgments with the model’s answer in view. Non-blind coding can only inflate agreement, which makes the low agreement estimates reported below conservative in the direction that matters. A second, fully blinded coder, with an overlap set for inter-rater reliability, is in progress.

Prosecutor attribution and relevance (Section 3.2). The most consequential audit finding concerns relevance. The research assistant judged 28 of the 100 articles in the stance-and-frame sample not substantively about the attributed prosecutor—typically articles that mention a district attorney in passing while covering something else. A simple relevance rule (at least two named mentions of the primary prosecutor, or a headline mention) separated these 28 articles from the other 72 with 100% recall and precision on this sample, and a continuous weighted centrality score achieves $AUC = 0.90$. This finding motivates the prosecutor-focused sensitivity analysis (Appendix F), in which the composite differential increases to $d = -0.260$, pointing the same way as the fallback-attribution sensitivity: peripheral articles dilute rather than drive the differential. The two are not independent evidence, however—the prosecutor-focused subsample is very nearly a subset of the non-fallback subsample (3,991 of its 3,999 articles), as it must be, since requiring two named mentions implies at least one. Notably, the research assistant flagged low-relevance or wrong-target articles independently in each of the first three packets, unprompted in the second and third, where relevance was not the assigned task: this is a consistent, coder-identified measurement problem rather than an isolated observation. Pooling those flags yields 55 flagged articles among the 333 distinct articles coded across those three packets, a benchmark substantially larger than the packet-01 sample on which the rule was built. The pooled benchmark and the rule grading below are reproducible: `ra_langextract_validation/scripts/build_relevance_benchmark.py`

rebuilds the flag set from the completed coding sheets under an explicit, documented note-matching rule and writes `generated/pooled_relevance_benchmark.csv`. Graded against it, the simple rule drops 70.9% of flagged articles (95% Wilson CI [57.9,81.2]) while 91.4% of the 185 articles it retains are genuinely prosecutor-focused, against 100% on both counts on the clean 28-article packet-01 benchmark. The residual failure mode is specific and worth naming: articles that name the attributed prosecutor several times while actually being about a different prosecutor—most often another reform prosecutor outside the study, with Gascón, Harris, or Rosen named in 19 of the 55 flagged articles. Adding a mention-dominance requirement—the primary prosecutor is named at least twice or in the headline *and* more often than all other prosecutors combined, including prominent non-study prosecutors—raises this to 83.6% of flagged articles dropped (CI [71.7,91.1]) at 94.5% precision on the 163 articles retained, while matching the simple rule’s perfect separation on the packet-01 benchmark. Under the stricter rule the composite differential is $d = -0.267$ ($p = 1.3 \times 10^{-16}$; Appendix F). One caveat is essential: the dominance rule was developed using these same flags, so its evaluation is in-sample, and out-of-sample confirmation with the second blinded coder is pending.

Evaluative stance (Method B, Section 4.1). The stance layer fared worst at the article level. Agreement between the human coder and the model’s discretized stance labels was no better than chance: Cohen’s $\kappa = 0.049$ (95% bootstrap CI [−0.08,0.18]), computed on the 84 of the 100 sampled articles for which the model produced a stance label at all—the remaining 16 contain no named prosecutor mention, so there is no stance score to discretize—with the classifier systematically over-assigning the supportive category (within those 84: 45 model-supportive articles versus 18 human-supportive). Recalibrating the discretization thresholds does not repair agreement. The continuous stance score, however, is monotonically ordered in the human labels: mean scores rise from +0.03 for human-labeled critical articles through +0.18 for neutral/mixed to +0.27 for supportive, and the continuous score separates human-supportive from human-critical articles with AUC = 0.69. Because the stance channel carries the composite differential (Section 4.5), this is the validation result that most constrains how the headline contrasts may be read.

Narrative framing (Section 4.6). Dominant-frame agreement is weak but above chance ($\kappa = 0.173$, 95% CI [0.02, 0.33], $n = 71$; the coder left the frame field blank on 24 articles and the model assigned no dominant frame on 5 of the rest). Because the framing contrasts reported in Section 4.6 rest on the model-classified subset, this agreement estimate qualifies the frame assignment underlying them: the between-group differences in dominant frame are measured with an instrument that agrees only weakly with a human coder at the article level.

Case-type composition (Section 4.5). A composition confound was tested directly. Because progressive prosecutors are identified with policy reform, a natural alternative explanation for the composite differential is genre rather than evaluation: progressive coverage might be disproportionately policy and political commentary, which carries more evaluative content, while traditional coverage is disproportionately individual-case reporting. The research assistant coded a 120-article stratified sample—30 articles in each of the four prosecutor-type $\times$ sampling-bucket sampling strata—for whether each article is centered on a specific criminal case or is primarily policy and political coverage. The coding is internally consistent: the binary case-centered judgment agrees with a separate specific-case-present indicator on 112 of 120 articles (93.3%), and seven of the eight mismatches run in one direction, boundary cases in which a specific case appears but functions as context for a policy argument. The coder flagged these explicitly as judgment calls, and an explicit adjudication rule is being applied to them. The composition test returns a clear null: 24 of 60 progressive-coverage articles and 25 of 59 traditional-coverage articles are case-centered (Fisher exact $p = .853$; odds ratio 0.91 for case-centered coverage, progressive versus traditional; one row is excluded from the cross-tabulation because a spreadsheet column shift corrupted its prosecutor-type field). Case-type composition therefore does not differ detectably by prosecutor type, which rules out the composition confound: the composite differential is not attributable to progressive prosecutors receiving proportionally more policy coverage and traditional prosecutors more individual-case coverage. Within-case-type contrasts, however, cannot be estimated from this sample. The subgroup cells such a contrast would rest on are smaller and unequal—24 case-centered and 36 policy-centered progressive articles against 25 and 34 traditional—and at 24 to 36

articles per subgroup the design has roughly 0.2 power against the corpus effect size. The subgroup nulls are therefore uninformative about whether the differential holds within case type and are not reported here as evidence.

Structural extraction, precision (Section 4.8). The research assistant reviewed 160 individual extractions from the extraction layer (Appendix A)—60 causal claims, 86 source attributions, 9 policy actions, and 5 comparisons—verifying for each whether the extracted span appears in the article, whether the extraction class is correct, and whether the structured attributes are correct. The sample was deliberately adversarial rather than corpus-representative: 40 of the 160 extractions were drawn from a schema-drift bucket sampled precisely because they looked suspicious, and the remainder were half purposively selected, so the precision rates below are conservative lower bounds, not corpus-level estimates. Span grounding is near-perfect: the research assistant initially marked 26 extractions as absent from the provided excerpt, but 24 of these appear verbatim in the full article text—an artifact of packet construction rather than model error—leaving 158 of 160 extractions textually grounded (98.8%, 95% Wilson CI [95.6, 99.7]), with the remaining two possible paraphrases. Class assignment is likewise highly precise: 142 of 147 judged extractions carry the correct class (96.6%, CI [92.3, 98.5]), including 83 of 83 source attributions (100%). The weak layer is the structured attributes: 105 of 145 judged extractions have fully correct attributes (72.4%, CI [64.6, 79.0]), with source attributions at 84.3% (70/83, CI [75.0, 90.6]) but causal claims at only 62.5% (30/48, CI [48.4, 74.8]); policy actions (3/9) and comparisons (2/5) have too few audited cases for stable estimates. The error taxonomy from the research assistant’s corrections identifies the failure modes: seven causal claims about *other* prosecutors mentioned in the article (Gascón, Spitzer, Chisholm) were attributed to the article’s primary prosecutor (cross-prosecutor contamination); twelve causal claims over-attributed responsibility to the prosecutor where the text blames other actors or institutions; four source attributions confused the speaker’s role (e.g., expert coded as other, prosecutor coded as defense attorney); and eight source-stance labels were corrected, most commonly model “critical” revised to human “neutral” (six cases).

Because the harm-attribution differential reported in Appendix A rests on the causal-direction attribute, the audit’s downgrade pattern matters. Of the 41 model-assigned “prosecutor caused harm” labels in the audited set (33 in progressive-prosecutor coverage, 8 in traditional—mirroring the corpus differential), the research assistant downgraded 6 of 33 progressive labels (18%, CI [9, 34]) versus 1 of 8 traditional labels (12%, CI [2, 47]) to “ambiguous.” There is thus no evidence of asymmetric error that would manufacture the harm-attribution differential, but roughly one in six harm labels does not survive human review: attribute-level ratios (harm attribution, source stance) carry materially more measurement noise than the class-level counts, whose precision is 96.6% pooled and 100% for source attributions.

Structural extraction, recall. A separate packet measured extraction *recall*: content present in an article that the model failed to extract. It is the only packet in the program that quantifies false negatives, since a precision audit cannot by construction detect what was never extracted. The research assistant was shown an article excerpt together with the model’s extraction list for that article, and counted the attributed sources and causal claims present in the excerpt that the model had not captured, across 40 articles (20 progressive, 20 traditional). She recorded 80 missed source attributions and 9 missed causal claims, with 30 of the 40 articles containing at least one missed source. Ninety percent of all missed items are source attributions: the model’s recall problem is concentrated almost entirely in source capture, not causal-claim capture. The false-negative rate for sources is best reported as a range, 28–34%, because the denominator is ambiguous. Treating every model extraction for these articles as a recall success gives 27.7% (80 missed against 209 extracted); counting only those extractions verifiably located inside the excerpt the coder could actually see gives 33.6% (80 of 238; 95% Wilson CI [27.9, 39.8]). The truth lies between the two bounds, because the model can legitimately capture a source through a quotation that falls outside the displayed excerpt, yet such an extraction cannot fairly count as a success against a judgment the coder formed on the excerpt alone. The corresponding range for causal claims is 31–39% (naive 31.0%, corrected 39.1%, CI [22.2, 59.2]), which rests on only nine missed items and is correspondingly imprecise.

The load-bearing question, however, is not the level of recall but its symmetry, because only *differential* recall failure could manufacture the source-ecology differential reported in Appendix A. Missed sources average 1.55 per article in progressive-prosecutor coverage ($n = 20$) versus 2.45 in traditional coverage ($n = 20$), Welch $p = .310$; missed causal claims average 0.15 versus 0.30, $p = .268$. There is no detectable asymmetry in recall failure across prosecutor types, so differential recall failure does not appear to manufacture the source-ecology differential. One qualifier is owed: the point estimates run toward *more* missed sources in traditional coverage, a direction that, if real, would inflate rather than create the progressive-traditional gap, but with 20 articles per group this comparison is badly underpowered and should not be leaned on. What the model misses, by contrast, is systematic and worth naming. The missed items are dominated by unnamed and documentary attribution—“authorities,” “investigators,” “officials,” court records, jail records, court testimony, the motion, the suit, newspaper reports, the coroner—while named individuals who are directly quoted are reliably captured. Recall failure is therefore not random attrition but a skew away from institutional and documentary voices toward the named, quoted human source.

One packet-design defect qualifies the recall estimate itself, and the research assistant identified it independently. The extraction pipeline ran the model on the first roughly 3,000 words of each article, but the review packet displayed only the first eight paragraphs, capped at 6,000 characters. As a result 101 of the 359 listed extractions (28.1%) lay outside the text the coder could see. Every one of those 101 was subsequently verified as present in the full article text, so this is an artifact of packet construction rather than ungrounded extraction—the same pattern, and the same conclusion, as the grounding result reported above (98.8%). The packet builder has been corrected so that recall packets list only extractions grounded in the displayed excerpt. This defect is why the recall rates above are bracketed rather than point-estimated.

These results impose a specific interpretive discipline on the rest of the paper: the stance measure supports aggregate, relative group contrasts only, and neither article-level classification nor absolute stance levels are supported by the validation evidence (Limitation 4, Section 5.2).

5 Discussion

This study provides the first large-scale, multi-method analysis of media coverage of prosecutors by ideological orientation. Three principal findings emerge.

First, media bias toward prosecutors is multi-dimensional, and the dimensions diverge. The study’s central contribution is demonstrating that what is commonly treated as a unitary construct, “media bias,” decomposes into at least two independent dimensions: evaluative framing and emotional tone. Stance classification detects a medium effect ($d = -0.337$) that survives regression controls ($\beta = -0.128$, $p < .001$); sentiment analysis detects at most a trace effect in the opposite direction ($d = 0.047$; $\beta = 0.010$, $p = .507$ after controls), and both sentiment channels are formally equivalent to zero by per-method TOST tests at the $|d| < 0.2$ bound ($p < .001$). This divergence is not a methodological artifact but a substantive finding: the media covers progressive prosecutors in the same emotional register as their traditional counterparts but evaluates their performance more critically. This pattern aligns with Entman’s (1993) observation that framing operates through selection and salience rather than through overt evaluative language, and with prior scholarship documenting how structural accumulation of story selection, source amplification, and contextual juxtaposition creates critical impressions without lexical negativity (Beale, 2006; Surette, 2015). The prosecutor-attributed theme analysis illustrates the mechanism most directly: when themes must be explicitly linked to prosecutors through compound linguistic patterns, the raw effect size increases to $d = -0.418$, consistent with an evaluative differential that operates through the *attribution* of critical narratives rather than through diffuse tonal differences. That layer, however, will not bear the weight of a differential-treatment claim, for reasons developed under the second finding below. The full-corpus structural extraction converges with this result: progressive prosecutor coverage is organized around more externally contestatory source ecologies and more harm-attribution content, while traditional coverage relies more heavily on neutral self-quotation. Together, the score-based and extraction-based layers support a complementary interpretation of differential coverage while remaining descriptive rather than causal.

The Alameda County comparison illustrates why decomposition matters. A non-significant composite ($d = -0.069$, $p = .138$) conceals substantial stance and keyword differentials pulling against near-zero, oppositely signed sentiment effects (Section 4.2, Table 3). A single-index approach would miss the differential treatment entirely.

Second, evaluative bias is event-driven rather than stable. The per-method temporal analysis (Figure 5) resolves a key ambiguity. Segmented ITS estimates around transition dates (Appendix G) reinforce the same pattern: limited transition effects in San Francisco but clear immediate evaluative shifts in Alameda, concentrated in stance and keyword outcomes rather than sentiment; the composite-level ITS coefficients themselves do not survive family-wise multiplicity correction and are suggestive only. Stance classification and keyword analysis show dramatic quarterly variation ($SD = 0.32$ and 0.36 , respectively), with the deepest effects spanning the Boudin recall and its aftermath (2021Q4–2022Q4). The study’s second recall does not behave the same way. Across the Price recall quarters the keyword differential is among its shallowest post-2021 values, so of the two recall campaigns this corpus contains, one coincides with the largest evaluative differentials in the study and the other does not. The recall observation therefore rests on a single case, and with two instances the design cannot separate a recall mechanism from the particular prosecutor or media market in which it occurred. Aspect sentiment and document sentiment vary comparably in magnitude ($SD = 0.28$ and 0.22) but show no pattern tied to political events, their largest quarterly values falling in pre-tenure 2019 rather than around any transition or recall. The aggregate temporal heterogeneity (composite range $d = -0.38$ to $+0.86$, $SD = 0.39$) is driven entirely by the evaluative dimensions; tonal bias is absent throughout. What this design estimates across more than one instance is not recalls but transitions in office: the segmented ITS fits both changes of officeholder against an untreated San Mateo comparison series, and the stance effects at the twelve-month horizon survive family-wise correction in both counties (Benjamini–Hochberg adjusted $p = 2.2 \times 10^{-4}$ in each). The reading the temporal evidence supports is correspondingly narrower than a claim about recall campaigns: evaluative scrutiny of progressive prosecutors rises around changes of officeholder and the contested episodes surrounding them, rather than reflect-

ing a stable editorial disposition against them. Accordingly, the term “bias” as used throughout this paper denotes differential evaluative responsiveness to political events, not a stable editorial preference: the finding is not that newsrooms maintain a constant anti-progressive disposition, but that politically charged events trigger evaluative scrutiny of reform prosecutors that has no tonal counterpart in coverage of traditional prosecutors facing comparable circumstances. The theme decomposition (Section 4.7) is the sharpest available evidence for this reading, and it also marks its limit. Removing the recall and crime-rising themes removes the theme differential entirely and then reverses it, which means the thematic layer measures event responsiveness almost exclusively. The qualifier “facing comparable circumstances” is doing real work here: the traditional prosecutors in this study did not face recall campaigns, so for the theme layer there is no like-for-like comparison to be made, and the event-driven interpretation is not a softening of the finding but a description of what the finding is.

Third, framing differences are more pronounced and potentially more consequential than any tonal measure. The Cramér’s V of 0.282 for dominant frame distributions among model-classified articles ($V = 0.214$ in the pooled sample, where roughly half the rows are assigned by dictionary fallback), a medium effect, indicates that progressive and traditional prosecutors are covered through substantively different narrative lenses. Progressive prosecutors are held to an accountability standard (40% of articles vs. 19%) and viewed through an ideological lens (5% vs. 1%), while traditional prosecutors receive more human-interest framing (32% vs. 17%) and more conflict framing (48% vs. 37%). These framing choices may matter more than any aggregate tone measure: they determine whether voters receive performance evaluations or human-interest stories about structurally similar officeholders. One qualification travels with this finding and is developed in Section 4.6: the continuous frame-probability contrasts can only be read on the model-classified subset, because the pooled frame scores mix two instruments whose levels differ roughly thirtyfold and whose assignment is correlated with prosecutor type. The pooled probability contrasts published in an earlier version of this paper are withdrawn.

5.1 Researcher Degrees of Freedom and the Ordering of Effect Sizes

A pattern in this paper’s results is visible to any reader who lines the measures up, and it is better stated by the author than discovered by a referee. Rank the study’s instruments by how much of each one I constructed, and the effect sizes rank with them, nearly monotonically:

- **Off-the-shelf sentiment models** (Methods A and D), which I did not design and could not tune: $d = +0.047$ and $d = +0.049$, both reversed in sign and both TOST-equivalent to zero.
- **The LLM bias-indicator schema** (Appendix B), which I wrote but deliberately symmetrized—identical extraction standards for favorable and unfavorable indicators, fictional exemplars with matched extraction densities across slant directions: $d = -0.039$ on $n = 999$, also TOST-equivalent to zero.
- **Researcher-authored zero-shot hypothesis labels** (Method B, stance), where I wrote the label strings the classifier tests: $d = -0.337$.
- **A researcher-curated keyword dictionary** (Method C): $d = -0.203$.
- **Researcher-authored theme attribution** (Section 4.7), with hand-written compound patterns, hand-set method weights, and hand-set agreement multipliers: $d = -0.418$, the largest pooled effect in the paper in magnitude.

The ordering is not perfect—stance and keywords sit in the wrong order relative to how much of each I authored—but the shape is unmistakable: the more of the instrument I built, the larger the differential it reports.

Part of this conflates researcher input with *construct*, and that part is a genuine defense. The off-the-shelf sentiment models are not low-discretion measures of the same thing the constructed instruments measure; they are low-discretion measures of emotional tone, and this paper’s central claim is that emotional tone is the wrong construct for evaluative bias. Their nulls are a finding, not a benchmark that the constructed instruments failed to meet. There is no off-the-shelf instrument for “does this article evaluate this prosecutor critically” that could have been used instead, which is

the reason the stance, keyword, and theme layers were built at all. The bias-indicator schema is the informative middle case: it targets an evaluative construct, it was written by me, and it still returns a null—but it was symmetrized by design, which is exactly the discipline the other constructed instruments do not have.

The rest of the concern stands, and this paper’s own evidence now supports it. The most heavily constructed instrument in the study, theme attribution, is precisely the one shown in Section 4.7 to be substantially event-driven and to reverse sign once recall-flagged and crime-rising-flagged articles are removed. The instrument with the most researcher input produced both the largest effect and the least durable one. That is what a critic raising this objection would predict, and it is what happened. The implication for how to read the paper is not that the constructed measures are worthless—their signs are consistent across within-county pairs, regression controls, and transition-date models—but that their *magnitudes* should be discounted relative to their signs, and that the quantities to trust are the ones estimated with the least discretion available for a given construct: the two sentiment nulls, the symmetric bias-indicator null, and the within-county rather than pooled versions of the constructed contrasts.

5.2 Limitations

The limitations below are ordered by how much they constrain what the paper can claim, beginning with the two that bound the headline estimate and ending with matters of scope. Several were identified by the validation program itself (Section 4.9) rather than anticipated in advance, and those are the ones that changed the paper most.

1. **The design is observational, and the groups did not face comparable circumstances.** Differences in coverage may reflect genuine differences in prosecutorial conduct, public controversy, or newsworthy events rather than ideological bias per se. Boudin’s recall, Price’s contentious relationship with law enforcement, and the political dynamics of Bay Area criminal justice are confounds that within-county comparisons mitigate but do not eliminate. This is not a generic

caveat here: both progressive prosecutors in the study faced actual recall campaigns and no traditional prosecutor did, so for several outcomes there is no like-for-like comparison available at all.

2. **The magnitude of the pooled composite depends on the composition of the traditional baseline.** Across the five leave-one-prosecutor-out fits the composite ranges from $d = -0.233$ to $d = -0.070$ (Appendix E). The low end drops Steve Wagstaffe, whose coverage is the most favorable in the corpus and who is the only prosecutor from San Mateo County, the only county with no transition; his removal halves the pooled effect. The pooled contrast is thus partly a between-county comparison anchored on a single untreated county—the same county that serves as the comparison series in the controlled segmented ITS, so one county is load-bearing in two places with no second untreated county available to check either. The sign is robust across all five fits; the magnitude is not. The value $d \approx -0.07$, at which the two within-county pairs and the drop-Wagstaffe fit agree, is the better estimate, and the pooled $d = -0.151$ should be read as an upper bound.
3. **The theme differential is event-driven and does not survive as a differential-treatment finding.** The prosecutor-attributed theme effect is the largest raw effect in magnitude among the pooled contrasts ($d = -0.418$), but two themes carry most of it, and both track events that actually differed between the groups. Excluding articles flagged with either theme reverses the sign of the differential ($d = +0.088$; Section 4.7). The portion not attributable to these two themes is small and slightly favors the progressive group. The theme layer is evidence about *what* progressive-prosecutor coverage was about during these years, not an estimate of how comparable officials are framed. It is also the most heavily researcher-constructed instrument in the study, and the coincidence between that fact and its fragility is discussed directly in Section 5.1.
4. **Article-level stance classification does not agree with a human coder beyond chance.** Co-

hen's $\kappa = 0.049$ (95% CI $[-0.08, 0.18]$, $n = 84$), with the classifier systematically over-assigning the supportive category; recalibrating the discretization thresholds does not repair it (Section 4.9). The continuous score is monotonically ordered in the human labels (AUC = 0.69), and symmetric classification noise attenuates rather than inflates group contrasts, so the aggregate stance differential remains interpretable. But stance supports *aggregate*, *relative* contrasts only. No claim about an individual article, or about the absolute stance level of any prosecutor's coverage, is supported by the validation evidence, and the score's zero point is positively miscalibrated. Dominant-frame agreement is weak though above chance ($\kappa = 0.173$).

5. **The validation program rests on a single, non-blind coder, and the relevance rules are graded in-sample.** Every validation result reported in this paper derives from a single coder's labels, and each packet displayed the model's outputs alongside the article text, so the coding was not blind to those outputs. No inter-rater reliability estimate is reported anywhere in this paper: the κ values above measure human–model agreement, not human–human agreement, so there is at present no way to separate model error from coder idiosyncrasy in any of them. A second, fully blinded coder with an overlap set is in progress and will supply that estimate. Non-blind coding can only inflate human–model agreement, which makes the low agreement estimates above conservative, but it also means the audit's *favorable* findings—span grounding, class precision, the case-type composition null—rest on one person's judgment with no independent check. Separately, the mention-dominance relevance rule was developed against the same 55 flagged articles it is graded on, so its reported 83.6% recall at 94.5% precision is an in-sample upper bound, not a validated out-of-sample gain over the simpler rule.
6. **Extraction recall is the weak side of the ledger, and what it misses is systematic.** The recall audit puts the false-negative rate for source attributions at 28–34% and for causal claims at 31–39% (Section 4.9). Missed items are dominated by unnamed and documentary attribution—“authorities,” “investigators,” court and jail records, testimony, filings—while named, directly quoted individuals are reliably captured. The absolute source-composition shares in Appendix A

therefore understate institutional and documentary sourcing, and the chi-square on the overall source-type distribution should not be read as a claim about true composition. Between-group ratios are the more robust quantity, because recall failure shows no detectable asymmetry across prosecutor types, though that symmetry test is badly underpowered (Section 4.9). Attribute-level precision is likewise the weakest layer of the extraction (72.4% overall, 62.5% for causal claims), so attribute-based ratios such as harm attribution and source stance carry materially more noise than the class-level counts.

7. **The framing fallback and the attribution fallback are one subsample, not two.** The keyword frame fallback and the county-and-date attribution fallback flag almost exactly the same articles (Section 3.4). The primary (zero-shot-only) framing analysis and the exclude-fallback composite sensitivity are consequently two instruments applied to one subsample, and the paper does not treat their agreement as independent corroboration. It also means that roughly half of the analysis corpus is measured by dictionary rather than by model on the framing layer, and that the same half is attributed by county-and-date rather than by name. A direct consequence is that the primary framing estimates rest on half the corpus, and on the half in which prosecutors are named, so they are better measured but not representative of the whole attributed sample.
8. **The two frame instruments are not on a common scale, and pooling them produces figures that are not measurements.** The zero-shot classifier returns entailment probabilities and the keyword fallback returns dictionary match fractions; mean levels differ by roughly a factor of thirty, and instrument assignment is correlated with group (Section 3.4), so pooled means partly track which instrument scored an article. That this is fatal rather than merely noisy is established by Table 5: for conflict and human interest the pooled Cohen’s d lies outside the range spanned by its own two subgroups, which no weighted combination on a common scale can do. The continuous frame contrasts are therefore estimated on the model-classified subset only, and the pooled probability contrasts reported in an earlier version of this paper are withdrawn. The dominant-frame chi-square is less exposed, because argmax within an article is in-kind across

instruments, but the fallback’s argmax runs over near-zero values, so the pooled chi-square is reported as secondary to the model-classified one.

9. **Each instrument has known failure modes, and one of them is unchecked where it matters most.** Zero-shot classifiers can confuse quoted source stance with authorial stance in attribution-heavy journalism (e.g., reading “critics say Boudin is soft on crime” as the article’s own position); sentiment models trained on social media transfer imperfectly to news prose; and keyword dictionaries embed researcher assumptions and miss surface variants they do not enumerate (Appendix D). Quote-masked sensitivity variants exist for the keyword channel and the theme layer, and both leave the estimates essentially unchanged—but no quote-masked variant exists for the stance classifier, which is the method carrying the paper’s main evaluative result. The quoted-speech concern is therefore mitigated for the layers where it is least consequential and untested for the one where it is most.
10. **News production is redundant, so the effective sample is smaller than n suggests.** A single press conference or recall milestone generates near-duplicate stories across outlets and wire services. Cluster-robust standard errors by publication address within-outlet dependence but not story-level clustering across publications covering the same event. The extreme p -values reported throughout (e.g., $p < 10^{-123}$) should be read as strong directional evidence rather than precise probability statements; the effect sizes, less sensitive to sample-size inflation, are the substantively meaningful quantities.
11. **The corpus is one metropolitan area, and it is not purely local.** Whether these patterns generalize to reform prosecutors elsewhere (Krasner in Philadelphia, Foxx in Chicago, Gascón in Los Angeles) is an open empirical question. Within the corpus itself, two of the 21 publications are national rather than Bay Area outlets, supplying 3.1% of the analysis sample, and their coverage is selected on national newsworthiness in a way local coverage is not; the imbalance this produces is visible in their disproportionately progressive attributed coverage (77% against

51% overall; Section 3.1). At 3.1% of the analysis sample the two outlets cannot drive the reported contrasts, but they tilt in the direction of the reported effect rather than against it, and the paper’s framing as a study of *Bay Area* coverage is an approximation rather than an exact description of the corpus.

5.3 Implications and Future Directions

These findings have implications for scholarship on prosecutorial politics, media bias, and computational text analysis. For prosecutorial politics, the results suggest that reform prosecutors face a structurally different media environment: one organized around accountability and ideological evaluation rather than the human-interest narratives that characterize coverage of traditional prosecutors. This asymmetry may contribute to the political vulnerability of progressive prosecutors, who face constant performance evaluation in the press while their traditional counterparts benefit from more diffuse, story-driven coverage.

For media bias research, the sentiment/stance divergence is a measurement warning as much as a substantive finding: single-instrument sentiment designs would report equivalence here (Section 4.1).

These findings also raise concerns about democratic accountability in criminal justice. Media coverage is the primary conduit through which the public learns about prosecutorial policy (Wright, 2009), and the results indicate that this conduit applies systematically different evaluative standards depending on a prosecutor’s ideological orientation. Two constraints on that claim follow from the sensitivity analyses above and should travel with it. The evaluative differential is concentrated in framing and sourcing rather than tone, and its magnitude is smaller than the pooled composite headline suggests: the designs that avoid cross-county comparison converge on $d \approx -0.07$ rather than $d = -0.151$. And the thematic component of the differential is event-linked, not a standing standard applied to like cases; the recall coverage that dominates it responded to recall campaigns that only the progressive prosecutors faced. What the results support is that differen-

tial *evaluative framing and narrative selection* accompany reform prosecutors’ tenures, not that newsrooms hold reform prosecutors to a harsher standard on identical facts. When progressive prosecutors are disproportionately covered through accountability frames while traditional prosecutors receive human-interest coverage, voters receive different types of information about structurally similar officeholders, developing calibrated judgments about a reform prosecutor’s policy outcomes while remaining uninformed about a traditional prosecutor’s performance. This asymmetry is compounded by the “toplash” dynamics [Goldstein \(2024\)](#) describes, in which institutional actors leverage media relationships to amplify opposition to reform. If small, persistent shifts in evaluative framing accumulate over thousands of articles, they may contribute to the political vulnerability of reform prosecutors through a mechanism that is invisible in any individual article but consequential in aggregate. Taken together, these results demonstrate that the press covers reform prosecutors differently in kind rather than in degree, not through uniform hostility, but through systematic differences in evaluative framing, sourcing, and narrative structure that are invisible to tonal analysis alone.

Future work should expand the corpus to include other reform prosecutors nationally, complete the human validation program begun here (Section [4.9](#)), including the second blinded coder and inter-rater overlap set now in progress, a genuinely out-of-sample test of the mention-dominance relevance rule, which is currently graded against the same flags used to construct it, and a larger recall audit on a corrected packet design, since the completed 40-article audit establishes symmetry of recall failure across prosecutor types only imprecisely, and pursue causal identification strategies, such as regression discontinuity designs around close prosecutorial elections, to distinguish ideological bias from coverage differences attributable to newsworthy events.

Acknowledgments

Guanwen Wan conducted the human validation program reported in Section [4.9](#), coding four packets covering 373 distinct articles: article-level stance and dominant frame, a span-level precision

audit of the structural extraction, a case-type composition audit, and an extraction recall audit. The relevance criterion applied in the sample-composition sensitivity analyses (Appendix F) originated in her observation—raised unprompted in packets where relevance was not the assigned task—that a substantial share of attributed articles are not substantively about the attributed prosecutor. She also identified two defects in the analysis code, including the excerpt-construction error described in Section 4.9, which would otherwise have inflated the estimated extraction false-negative rate. She bears no responsibility for the interpretations offered here.

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Appendix A LLM-Based Structural Content Extraction

The multi-method analyses reported in Section 4 measure evaluative direction, narrative framing, and prosecutor-attributed themes but do not directly enumerate the structural building blocks of articles: who is quoted, what claims are made, and with what evidence. Understanding these structural features is essential for explaining *why* evaluative coverage differs across prosecutor types, yet manual content coding at the scale of $n = 12,827$ articles is prohibitively expensive. This appendix reports extraction protocol details and extended tables for the full-corpus LLM-based structured extraction.

Method

I use Google’s langextract library, which leverages Gemini 2.5 Flash to extract structured information from text with source grounding: every extraction is anchored to a specific span in the original article. The model was prompted with a detailed extraction schema and three diverse few-shot examples (one critical of a progressive prosecutor, one neutral about a traditional prosecutor, one with mixed coverage) to calibrate extraction behavior across ideological contexts. I applied the extraction pipeline to the full corpus of 12,827 prosecutor-attributed articles (6,585 progressive, 6,242 traditional), covering all five prosecutors: Boudin ($n = 5,783$), Jenkins ($n = 3,095$), O’Malley ($n = 1,495$), Price ($n = 802$), and Wagstaffe ($n = 1,646$). Six articles whose extraction runs returned processing errors are excluded from extraction-level statistics, so these per-prosecutor extraction counts total 12,821 rather than 12,827.

Extraction prompt

The following prompt was passed to the model for every article. It defines five extraction classes, each with structured attributes that constrain the model's output to a fixed schema:

Extract the following types of information from this news article about a district attorney / prosecutor. Use exact text spans from the article.

1. `claim_against_prosecutor`: Specific accusations, criticisms, or negative claims made about the prosecutor's performance, policies, or character.

Attributes: `claim_type` (performance | policy | character | competence), `specificity` (vague | specific | quantified), `evidence_cited` (none | anecdotal | statistical | official_report).

2. `source_attribution`: Identify who is speaking, quoted, or cited as a source.

Attributes: `source_type` (police | victim | defense_attorney | prosecutor | politician | community_member | journalist | expert | advocacy_group), `stance_toward_prosecutor` (critical | supportive | neutral).

3. `causal_claim`: Claims that the prosecutor's actions caused or contributed to some outcome.

Attributes: `effect` (crime_increase | public_safety_decline | case_outcome | community_impact | positive_outcome), `causal_strength` (explicit | implied | speculative), `direction` (prosecutor_caused_harm | prosecutor_helped | ambiguous).

4. `policy_action`: Concrete policies, decisions, or actions attributed to the prosecutor.

Attributes: `action_type` (declined_to_prosecute | reduced_charges | new_policy | fired_staff | reversed_predecessor | enhanced_prosecution | other), `domain` (drugs | property_crime | violent_crime | bail | sentencing | staffing | juvenile | general), `framing` (positive | negative | neutral).

5. `comparison`: Explicit comparisons between the current prosecutor and a predecessor, another prosecutor, or a general standard.

Attributes: compared_to (predecessor | other_prosecutor | general_standard), dimension (toughness | case_outcomes | policy | competence | ideology), who_favored (current | predecessor | neither).

Only extract items that are clearly present. Do not fabricate or infer. Extract text spans in order of appearance. Do not paraphrase.

Few-shot examples

Three examples were provided with each API call to demonstrate the expected extraction format. Each example pairs a synthetic article excerpt with the complete set of extractions the model should produce. The examples were designed to span the ideological and tonal range of the corpus.

Example 1: Critical coverage of a progressive prosecutor.

“San Francisco’s District Attorney Chesa Boudin faced renewed criticism this week after police data showed a 15% increase in car break-ins since he took office. Police Officers Association president Tony Montoya said officers are demoralized because suspects they arrest are quickly released without charges. ‘Why bother making arrests when the DA won’t prosecute?’ Montoya asked. Boudin defended his record, saying his office focuses on serious violent crime rather than low-level offenses. Recall organizers said the crime statistics prove Boudin’s progressive policies have failed San Francisco residents.”

Class	Extracted Text	Key Attributes
claim_against	“a 15% increase in car break-ins since he took office”	performance, quantified, statistical
source_attrib.	“Police Officers Association president Tony Montoya said officers are demoralized”	police, critical
source_attrib.	“Boudin defended his record. . .”	prosecutor, supportive
causal_claim	“the crime statistics prove Boudin’s progressive policies have failed”	crime_increase, explicit, harm
policy_action	“suspects they arrest are quickly released without charges”	declined_to_prosecute, negative
source_attrib.	“Recall organizers said. . .”	advocacy_group, critical

Example 2: Neutral coverage of a traditional prosecutor.

“Brooke Jenkins, who replaced Boudin after the recall, announced a new initiative targeting retail theft rings in the downtown area. Jenkins said her office would file felony charges against organized shoplifting groups, reversing Boudin’s practice of treating most retail theft as misdemeanors. Supervisor Matt Dorsey praised the move. Defense attorney Niki Solis warned that harsher penalties would disproportionately affect low-income communities without reducing crime.”

Class	Extracted Text	Key Attributes
policy_action	“file felony charges against organized shoplifting groups”	enhanced_prosecution, property_crime, positive
comparison	“reversing Boudin’s practice of treating most retail theft as misdemeanors”	predecessor, toughness, current
source_attrib.	“Supervisor Matt Dorsey praised the move”	politician, supportive
source_attrib.	“Defense attorney Niki Solis warned...”	defense_attorney, critical

Example 3: Mixed coverage of a progressive prosecutor.

“Alameda County DA Pamela Price faces a recall effort barely a year into her term. Critics say Price has been too lenient, pointing to cases where violent offenders received reduced sentences. The Save Alameda for Everyone committee cited three homicide cases where Price’s office offered plea deals. Price responded that her office has actually increased the conviction rate for violent felonies compared to her predecessor Nancy O’Malley.”

Class	Extracted Text	Key Attributes
claim_against	“Price has been too lenient”	policy, vague, none
claim_against	“violent offenders received reduced sentences”	performance, specific, anecdotal
source_attrib.	“The Save Alameda for Everyone committee cited three homicide cases”	advocacy_group, critical
policy_action	“Price’s office offered plea deals”	reduced_charges, violent_crime, negative
comparison	“increased the conviction rate... compared to her predecessor”	predecessor, case_outcomes, current
source_attrib.	“Price responded that her office has actually increased...”	prosecutor, supportive

The extraction schema defines five content categories: (1) *claims against the prosecutor*, classified by type (performance, policy, character, competence) and evidence quality; (2) *source attributions*, identifying the speaker’s role and stance toward the prosecutor; (3) *causal claims* linking the prosecutor’s actions to outcomes; (4) *policy actions* attributed to the prosecutor; and (5) *comparisons* between the current prosecutor and predecessors or standards. The pipeline produced 134,604 total extraction instances across 12,827 articles (mean = 10.5 per article).

Results

Source type distribution. The types of sources quoted differ significantly between prosecutor groups ($\chi^2 = 1802.1$, $p \approx 0$, $df = 9$). Main-text summaries are in Section 4.8; Table 7 provides the full source-type breakdown.

Coverage of progressive prosecutors features proportionally more advocacy group voices (2.4×), journalist commentary (1.5×), politician quotes (1.5×), and expert sources (1.4× per article). No-

Table 7: Source type distribution by prosecutor ideology. Proportions are within-group; rates are per attributed article (progressive $n = 6,585$; traditional $n = 6,242$).

Source Type	Prog. %	Trad. %	Prog./art.	Trad./art.	Ratio
Police	11.7%	11.7%	0.74	0.71	1.0×
Victim	2.5%	2.4%	0.15	0.15	1.0×
Defense attorney	6.7%	9.3%	0.42	0.57	0.7×
Prosecutor (self)	33.1%	42.3%	2.09	2.59	0.8×
Politician	10.0%	6.9%	0.63	0.42	1.5×
Community member	12.3%	11.2%	0.78	0.68	1.1×
Expert	7.7%	5.8%	0.48	0.36	1.4×
Journalist	7.2%	4.9%	0.46	0.30	1.5×
Advocacy group	6.9%	3.0%	0.44	0.18	2.4×
Other/missing type	1.9%	2.6%	0.12	0.16	0.8×

tably, traditional prosecutors receive more defense-attorney ($0.7\times$) and self-quote ($0.8\times$) sources per article, while police, victim, and community sources appear at roughly equal rates. These sourcing differentials suggest that progressive prosecutor coverage draws on a broader and more politically diverse source ecology, while traditional prosecutor coverage relies more heavily on the prosecutor’s own statements and legal actors. This is consistent with the accountability and reform framing patterns documented in Section 4.6, and with the “toplash” dynamics described by Goldstein (2024): the elevated presence of advocacy, politician, and expert sources around progressive prosecutors is precisely the source ecology one would expect when institutional actors mobilize media relationships to contest reform agendas.

The recall audit (Section 4.9) qualifies this table in a specific and asymmetric way. Because recall failure is concentrated in unnamed and documentary attribution, the model’s source inventory is biased toward named, directly quoted human sources. The *absolute* composition shares in the two percentage columns therefore understate documentary and institutional attribution, and the chi-square test of the overall source-type distribution should not be read as a claim about the true composition of either corpus. The *between-group ratios* in the final column, which are what the analysis actually relies on, are considerably more robust: the validation finds no detectable asymmetry in recall failure across prosecutor types (Section 4.9), so the omissions largely divide

out of a progressive-to-traditional ratio even though they do not divide out of a within-group share. The same qualification applies to the per-source-type rates: they are lower bounds on true sourcing density, and are interpretable comparatively rather than absolutely.

Source stance details are reported in the main text (Table 6); briefly, progressive prosecutor coverage has higher critical and supportive source rates, while traditional coverage has higher neutral source rates, largely driven by self-quotes (2.59 vs. 2.09 per article).

Causal claims. Progressive prosecutors face $1.7\times$ as many “prosecutor caused harm” attributions per article (0.42 vs. 0.24). “Prosecutor helped” attributions are also $1.2\times$ higher per article for progressives (0.15 vs. 0.13). The asymmetric harm ratio identifies a specific structural mechanism through which progressive prosecutors receive more negative coverage: causal blame is attributed more frequently regardless of evidence quality. The extraction audit qualifies this ratio: roughly one in six “prosecutor caused harm” labels did not survive human review, with no asymmetry across groups, so this attribute-based ratio carries materially more measurement noise than the source-ecology counts above (Section 4.9).

Claim types. Per article, progressive prosecutors face more claims across all categories, but the differentials are sharply asymmetric by type. Policy (0.49 vs. 0.31, $1.6\times$) and performance (0.45 vs. 0.31, $1.5\times$) criticisms show the largest gaps, while character (0.42 vs. 0.36, $1.2\times$) and competence (0.31 vs. 0.29, $1.1\times$) claims are relatively symmetric across ideology.

Per-prosecutor extraction density. Pamela Price generates the most extraction-dense coverage (13.6 per article, $n = 802$), followed by Brooke Jenkins (10.9, $n = 3,095$), Chesa Boudin (10.7, $n = 5,783$), Nancy O’Malley (10.0, $n = 1,495$), and Steve Wagstaffe (7.9, $n = 1,646$).²

²One might assume that progressive-prosecutor articles are simply richer or more detailed, producing more extractable content. The density data do not support this: a traditional prosecutor (Jenkins, 10.9) produces slightly *more* extractable content per article than the most-covered progressive (Boudin, 10.7), and the three middle prosecutors cluster between 10.0 and 10.9 regardless of ideology. The sourcing and claim differentials reported above reflect differences in *what kind* of content articles contain, not in how much.

Limitations

The extraction covers the full corpus of prosecutor-attributed articles ($n = 6,585$ progressive, $n = 6,242$ traditional), eliminating sampling bias as a concern. However, the LLM-based extraction itself introduces measurement uncertainty: Gemini 2.5 Flash may systematically over- or under-extract certain categories, and extraction quality likely varies with article length and complexity. A human validation program (Section 4.9) now covers article-level labels (relevance, stance, dominant frame), a span-level precision audit of 160 extractions, and a 40-article recall audit. The precision audit finds spans almost always textually grounded and extraction classes highly precise, but structured attributes correct in under three quarters of audited cases (Section 4.9). The class-level counts underlying this appendix’s source-ecology and claim-volume results are therefore robust to extraction error, while attribute-based ratios (harm attribution, source stance) should be read as directionally informative but noisier. Extraction *recall* is the weaker side of the ledger (28–34% of source attributions and 31–39% of causal claims missed), but it is statistically symmetric across prosecutor types, which is the property the between-group comparisons in this appendix require (Section 4.9). The source type asymmetry, particularly the advocacy-group disparity ($2.4\times$), suggests that tone differences detected in the main analysis may partly reflect differential journalistic sourcing practices rather than editorial bias per se.

Appendix B LLM-Based Bias Indicator Extraction

The structural extraction in Appendix A characterizes *what* articles contain; this appendix reports a complementary analysis that asks *how* the coverage is biased. A well-sourced, evidence-rich article can be legitimately critical of a prosecutor without being biased, while a superficially neutral article can exhibit bias through ungrounded assertions, source imbalance, loaded language, or omission of relevant context. Because the extraction schema is symmetric by construction—favorable and unfavorable indicators are extracted under identical standards—and the stratified sample is now large ($n = 999$), the analysis can affirmatively *bound* indicator-level asymmetry rather than merely

fail to detect it.

Method

Using the same langextract + Gemini 2.5 Flash infrastructure, I designed a schema targeting four categories of bias indicator: (1) *ungrounded negative claims*; (2) *source prominence imbalance*; (3) *loaded language*; and (4) *missing context*. A key methodological challenge is preventing the model from conflating legitimate criticism with bias, addressed through three calibration examples demonstrating that criticism with evidence is not bias. Following an initial pilot run, the schema was revised in two ways: the loaded-language class now carries a required *valence* attribute (favorable vs. unfavorable), with only unfavorable items counting toward the anti-prosecutor signal (favorable items feed a mirrored pro-prosecutor term); and the few-shot exemplars were re-written to feature fictional prosecutors with balanced extraction densities across slant directions, avoiding entity-keyed priors about the study prosecutors. All results reported below reflect extraction under this revised schema, applied to a balanced-by-type stratified sample of 1,000 articles drawn with a fixed random seed (42). The sampling is top-up aware: the 200 articles from the original pilot draw retain their prior extractions, and 800 newly drawn articles were extracted under the identical schema. One article failed with an API error and is excluded, leaving $n = 999$ analyzed (500 progressive, 499 traditional).

Extraction prompt

The following prompt was passed to the model for every article. Unlike the structural extraction in Appendix A, this schema explicitly distinguishes legitimate criticism from bias through calibration guidance:

```
Analyze this news article about a district attorney / prosecutor for BIAS
INDICATORS -- patterns of coverage that reveal systematic slant rather than
fair reporting. Use exact text spans from the article.
```

Only extract genuine bias indicators. Legitimate critical reporting with evidence is NOT bias. Balanced articles may have zero extractions.

1. `ungrounded_negative_claim`: A negative assertion about the prosecutor that lacks adequate supporting evidence within the article.

Attributes: `claim_content` (performance | policy | character | competence), `evidence_quality` (none | anonymous_source | single_anecdote | stats_without_baseline | adequately_sourced), `systemic_blame` (true | false).

2. `source_prominence_imbalance`: A source given disproportionate prominence relative to other viewpoints, creating an imbalanced picture.

Attributes: `source_stance` (critical | supportive), `prominence_mechanism` (placement_lede | placement_closing | extended_quote | sole_named_source | no_counterbalance | headline_framing), `counterbalance_present` (true | false).

3. `loaded_language`: Emotionally charged, non-neutral word choices that go beyond factual reporting.

Attributes: `language_type` (pejorative_label | scare_quotes | presuppositional_verb | hyperbole | ideological_framing | dehumanizing), `target` (prosecutor | prosecutor_policy | prosecutor_supporters | other), `valence` (favorable | unfavorable; required—does the loaded language cast its target in a favorable or unfavorable light?), `position` (headline_or_lede | body | closing).

4. `missing_context`: Important context that is conspicuously absent, making the coverage misleading.

Attributes: `what_is_missing` (trend_context | comparison_baseline | policy_rationale | systemic_factors | legal_constraints | prosecutor_response | alternative_explanation), `claim_it_supports` (free text).

CALIBRATION GUIDANCE:

- A well-sourced article that is critical of a prosecutor is NOT biased if

the criticism is grounded in evidence and the prosecutor’s perspective is included.

- An article that is favorable to a prosecutor IS biased if it cherry-picks achievements, omits failures, or relies on uncritical source selection.
- Apply the same standard regardless of which prosecutor is covered.

Only extract items that are clearly present. Do not fabricate or infer. Extract text spans in order of appearance. Do not paraphrase.

Few-shot examples

Three examples were provided to calibrate the model’s threshold for flagging bias. All prosecutors, counties, and sources in the exemplars are fictional, so the model brings no priors about the real study prosecutors to the extraction, and the anti-prosecutor and pro-prosecutor exemplars carry identical extraction counts (five each) so that neither slant direction is implicitly taught to be denser. Critically, Example 2 demonstrates that a well-sourced critical article should produce *minimal* bias indicators, preventing the model from conflating criticism with bias.

Example 1: Heavily biased anti-prosecutor article.

“Kern Valley County’s failed experiment: How DA Marcus Webb let criminals run wild. Since the prosecutor took office, residents say downtown Rivertown has become unrecognizable. Shoplifters brazenly clear shelves with impunity, and violent offenders walk free hours after arrest. ‘He cares more about criminals than victims,’ said retired officer Dan Kowalski, who spent 30 years on the force. Kowalski described case after case of suspects released without charges. Critics point to a 9% rise in property crime. Webb’s office did not respond to requests for comment by press time.”

Class	Extracted Text	Key Attributes
loaded_lang.	“Kern Valley County’s failed experiment”	pejorative_label, policy, unfavorable, headline
loaded_lang.	“let criminals run wild”	hyperbole, prosecutor, unfavorable, headline
ungrounded	“Shoplifters brazenly clear shelves with impunity, and violent offenders walk free hours after arrest”	performance, none, systemic_blame
source_imbal.	“retired officer Dan Kowalski...case after case of suspects released without charges”	critical, extended_quote, no counterbalance
missing_ctx.	“Critics point to a 9% rise in property crime”	comparison_baseline

Example 2: Well-constructed critical article (minimal bias).

“Sierra Madre County DA Elena Reyes’s first year in office has drawn sharp scrutiny. A Sierra Madre Herald analysis of court records found that Reyes’s office offered plea deals in 71% of violent felony cases, compared to 55% under predecessor Frank Delgado. The analysis examined 380 cases filed between January and September. Victims’ rights advocate Karen Liu said the plea rates concern families who expected tougher prosecution. Reyes defended the approach in a press conference, saying her office prioritizes cases with the strongest evidence and that conviction rates on cases taken to trial have actually increased to 88%. Criminal justice professor Alan Motravec at Ridgefield University noted that higher plea rates do not necessarily indicate leniency, as they may reflect more realistic case assessment. The recall campaign against Reyes has seized on the plea statistics.”

Class	Extracted Text	Key Attributes
missing_ctx.	“The recall campaign against Reyes has seized on the plea statistics”	systemic_factors (plea rates attributable to case composition, not ideology alone)

This example is critical for calibration: the article contains sharp criticism grounded in court records, includes the prosecutor’s response and an expert perspective, and produces only one extraction (a minor missing-context flag). This teaches the model that evidence-based critical reporting is *not* bias.

Example 3: Pro-prosecutor puff piece.

“New DA Susan Hale is restoring order to Port Landry County. In her first six months, the tough-on-crime prosecutor has brought a sense of accountability that residents desperately needed. Hale announced felony charges against a prolific shoplifter, drawing praise from the business community. ‘Finally, someone who takes our concerns seriously,’ said Harbor District Alliance director Tom Boyd, who described Hale as a breath of fresh air after the chaos of her predecessor’s tenure. Police Chief Rita Alvarez lauded the improved cooperation between her department and the DA’s office. Crime data for the period is not yet available from official sources.”

Class	Extracted Text	Key Attributes
loaded_lang.	“restoring order to Port Landry County”	presuppositional_verb, prosecutor, favorable, headline
loaded_lang.	“the tough-on-crime prosecutor has brought...accountability that residents desperately needed”	ideological_framing, prosecutor, favorable, body
source_imbal.	“Tom Boyd...a breath of fresh air after the chaos of her predecessor’s tenure”	supportive, extended_quote, no counterbalance
source_imbal.	“Police Chief Rita Alvarez lauded the improved cooperation”	supportive, no_counterbalance
missing_ctx.	“Crime data for the period is not yet available”	trend_context (effectiveness claimed without evidence)

This example demonstrates that the schema applies symmetrically: a favorable article triggers loaded-language (with favorable valence, feeding the mirrored pro-prosecutor term), source-imbalance, and missing-context indicators just as readily as a hostile one.

Results

Primary test. Progressive prosecutors receive a mean bias score of $M = -0.086$ ($n = 500$) versus $M = -0.076$ for traditional prosecutors ($n = 499$); the difference is small and not statistically significant: Welch’s $t = -0.61$, $p = .541$; Mann–Whitney $p = .233$; $d = -0.039$; bootstrap 95% CI for the mean difference $[-0.044, +0.023]$. Unlike the 200-article pilot, the sample is now large enough for the equivalence test to be informative: a TOST at $\delta = 0.2$ establishes equivalence ($p = .0055$), affirmatively bounding any aggregate indicator-level asymmetry within $|d| < 0.2$. The equivalence result contrasts with the significant $d = -0.151$ composite differential from the main pipeline (Section 4.5).

Within-county paired comparisons. The two same-county contrasts diverge. In San Francisco, Jenkins’s coverage is scored slightly more negatively than Boudin’s ($M = -0.090$, $n = 264$ vs. $M = -0.066$, $n = 434$; $d = +0.087$, $p = .264$), a non-significant difference. In Alameda, by contrast, Price’s coverage carries substantially more anti-prosecutor bias indicators than O’Malley’s ($M = -0.219$, $n = 66$ vs. $M = -0.093$, $n = 119$; $d = -0.433$, $p = .0095$). The overall equivalence thus masks offsetting county-level patterns; the Alameda asymmetry coheres with the composite, stance, and controlled-ITS results, which likewise locate the robust effects in Alameda. This is one of two paired tests and rests on modest cell sizes ($n = 66$ and $n = 119$), so it warrants correspondingly cautious interpretation.

Per-indicator decomposition. Individual bias indicators show small effects, most leaning toward higher counts in progressive-prosecutor coverage (Table 8; see also Figure 12); positive d indicates a higher count in progressive coverage, and no indicator-level significance tests are computed:

Table 8: Bias indicator means by prosecutor ideology (Appendix B, $n = 999$).

Indicator	Progressive M	Traditional M	d
Ungrounded claims	0.48	0.39	+0.07
Ungrounded (severe)	0.40	0.31	+0.09
Systemic blame	0.14	0.13	+0.02
Source imbalance (critical)	0.60	0.63	−0.03
Source imbalance (supportive)	0.25	0.20	+0.08
Loaded language (total)	1.45	1.00	+0.17
Loaded language (unfavorable)	0.54	0.36	+0.13
Loaded language (favorable)	0.17	0.12	+0.07
Loaded language (headline)	0.19	0.14	+0.10
Missing context	0.91	0.81	+0.08
Total indicators	3.70	3.04	+0.12

Loaded *unfavorable* language, the indicator most directly tied to the anti-prosecutor hypothesis, shows a modest differential at this sample size (0.54 vs. 0.36 per article, $d = +0.13$), and total loaded language is the largest per-indicator effect ($d = +0.17$). Critical-source prominence imbalance, the largest per-indicator effect in the 200-article pilot ($d = -0.24$, higher for traditional prosecutors), shrinks to near zero at $n = 999$ ($d = -0.03$). The remaining differentials are

small and mostly positive, including loaded *favorable* language ($d = +0.07$; progressive prosecutors receive more favorable loaded language, which feeds the mirrored pro-prosecutor term) and supportive-source imbalance ($d = +0.08$). The distribution of missing-context categories does not differ significantly across groups ($\chi^2 = 7.29$, $df = 7$, $p = .399$). Prominence-mechanism counts are broadly similar across groups (extended quotes: 244 vs. 249 instances; “no counterbalance”: 81 vs. 88), with progressive coverage showing more sole-named-source patterns (65 vs. 43).

Language type decomposition. Disaggregating loaded language by type, progressive prosecutor coverage exhibits more instances in every major category: ideological framing (230 vs. 152 instances), pejorative labels (204 vs. 122), hyperbole (163 vs. 124), presuppositional verbs (86 vs. 76), and scare quotes (39 vs. 20). The *composition* of loaded language does not differ significantly across groups ($\chi^2 = 9.00$, $df = 6$, $p = .174$): the differential is one of volume, not of type mix. Disaggregating by valence, progressive coverage carries more loaded language in both directions — unfavorable (605 vs. 418 instances) and favorable (122 vs. 81). Evidence quality for ungrounded claims is similar across groups: claims with no supporting evidence at all number 160 (progressive) vs. 121 (traditional).

Per-prosecutor decomposition. The most indicator-laden coverage in the sample belongs to Pamela Price (mean bias score -0.219 , $n = 66$), whose articles average 5.92 total indicators, including 1.36 critical-source imbalance flags, 1.18 loaded-unfavorable instances, 1.53 missing-context instances, and 0.91 ungrounded claims per article. Brooke Jenkins (-0.090 , $n = 264$) and Nancy O’Malley (-0.093 , $n = 119$) receive nearly identically scored coverage; Chesa Boudin’s is somewhat cleaner (-0.066 , $n = 434$); and Steve Wagstaffe receives the cleanest coverage of any prosecutor (1.59 total indicators and 0.19 ungrounded claims per article; mean bias score -0.024). The 200-article pilot’s most striking pattern — Jenkins as the most negatively scored prosecutor — does not persist at scale: her mean sits near the sample average, while Price’s stands clearly apart. Jenkins’s coverage remains indicator-dense in specific channels (0.78 critical-source flags per article vs. Boudin’s 0.48), consistent with the politically charged nature of her appointment,

but this heterogeneity now points primarily to Alameda: the aggregate equivalence result conceals substantial within-group variation concentrated in Price’s coverage. The most negatively scored prosecutor is thus a progressive, by a wide margin, while the two traditional prosecutors sit effectively tied just below the sample midpoint.

Convergent validity. The bias score correlates moderately with the aggregate tone-evaluation index from the main pipeline ($n = 999$ matched articles): Pearson $r = 0.245$, $p < .001$; Spearman $\rho = 0.251$, $p < .001$. The two approaches capture overlapping but distinct constructs: the NLP pipeline measures aggregate tone and evaluative stance, while the LLM extraction targets specific journalistic violations.

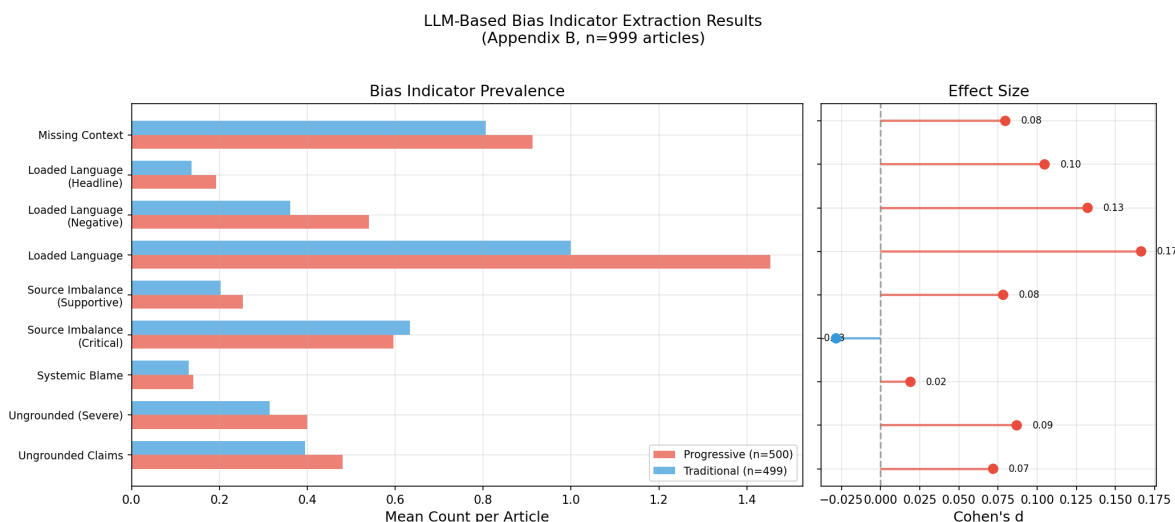


Figure 12: LLM-based bias indicator extraction results (Appendix B, $n = 999$). Left panel: mean indicator counts by ideology. Right panel: Cohen’s d effect sizes for each indicator. Loaded unfavorable language shows a modest differential toward progressive coverage ($d = +0.13$); the largest per-indicator effect is total loaded language ($d = +0.17$), while critical-source imbalance is near zero ($d = -0.03$).

Discussion

The primary result is an affirmative equivalence rather than a mere null. The interpretation is that media bias toward progressive prosecutors operates through tone and framing, the aggregate evaluative register of coverage, rather than through discrete, identifiable journalistic violations

(ungrounded claims, source imbalance, loaded language, missing context); at this sample size, that claim is a powered bound rather than a suggestive failure to detect.

An earlier version of this analysis, run before the valence revision, appeared to show a loaded-negative-language differential against progressive prosecutors and was read as a mechanistic bridge between the stance classifier’s evaluative signal and a specific lexical practice. The corrected schema complicates that story rather than simply erasing it. In the 200-article re-extraction the differential appeared to vanish ($d = +0.01$); at $n = 999$ a modest loaded-unfavorable differential re-emerges (0.54 vs. 0.36 per article, $d = +0.13$), alongside a smaller loaded-favorable differential ($d = +0.07$). These count differentials, however, do not produce net signed-score asymmetry — the primary test is TOST-equivalent — because the anti- and pro-prosecutor channels largely offset within the signed score. The strong mechanistic-bridge claim therefore remains withdrawn: progressive-prosecutor coverage contains somewhat more loaded language in *both* directions, a volume difference rather than a directional lexical mechanism for the stance effect detected in Section 4.1. The moderate convergent validity ($r = 0.245$) between the two pipelines indicates that the bias score is tone-adjacent but measures a substantially distinct construct.

The within-county comparisons sharpen this picture: discrete journalistic violations are not uniformly absent from the differential; they concentrate in Alameda, where the other instruments also detect it. At the same time, the per-prosecutor decomposition shows that prosecutors of both types attract indicator-dense coverage during politically contested periods (Jenkins’s critical-source flags remain elevated), reinforcing the main paper’s argument that bias detection requires multi-dimensional measurement: an approach sensitive only to explicit violations would miss the evaluative framing that drives the main effects, while an approach sensitive only to aggregate framing would miss the Alameda indicator asymmetry.

The analysis was upsized from a 200-article pilot to a stratified sample of 1,000 articles precisely to make the equivalence test and subgroup comparisons informative. The primary test is now well powered, though the Alameda paired comparison still rests on modest cells (up from 11 and

28 in the pilot). All results reported in this appendix reflect the revised valence-aware schema and neutralized fictional-prosecutor exemplars described in the Method section; the original pilot schema, which lacked the valence attribute and used entity-keyed exemplars, produced results that did not survive the correction and are superseded.

Appendix C Supplementary Diagnostic Figures and Tables

This appendix consolidates diagnostic visuals and supporting tables moved from the main text for readability.

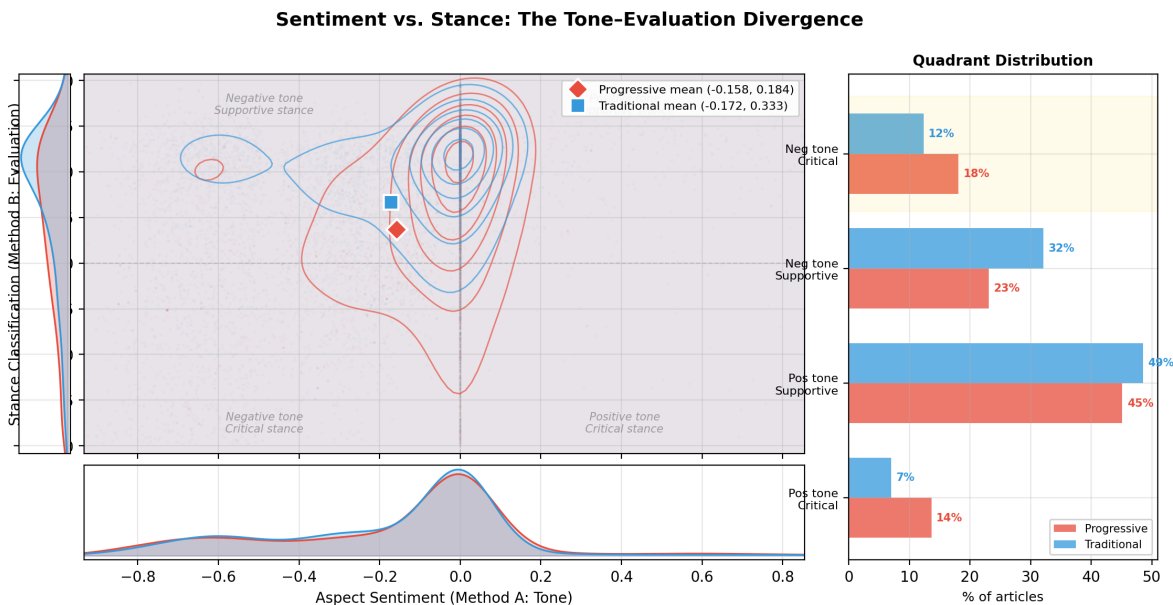


Figure 13: Tone–evaluation divergence at the article level. Central panel: KDE contour plot of Method A (aspect sentiment) vs. Method B (stance classification) scores for progressive (red) and traditional (blue) prosecutors. The horizontal separation in stance contrasts with the overlapping sentiment distributions, confirming that media coverage differs in evaluative framing rather than emotional tone.

Table 9: Per-method OLS regressions: progressive prosecutor coefficient with cluster-robust SEs.

Method	β	SE	p	95% CI	R^2
B: Stance	− 0.128	0.030	< .001	[−0.187, −0.068]	.073
C: Keywords	− 0.010	0.002	< .001	[−0.014, −0.006]	.053
A: Aspect sentiment	0.010	0.016	.507	[−0.020, 0.041]	.010
D: Document sentiment	0.017	0.013	.195	[−0.009, 0.043]	.006
Composite	−0.021	0.013	.104	[−0.045, 0.004]	.023

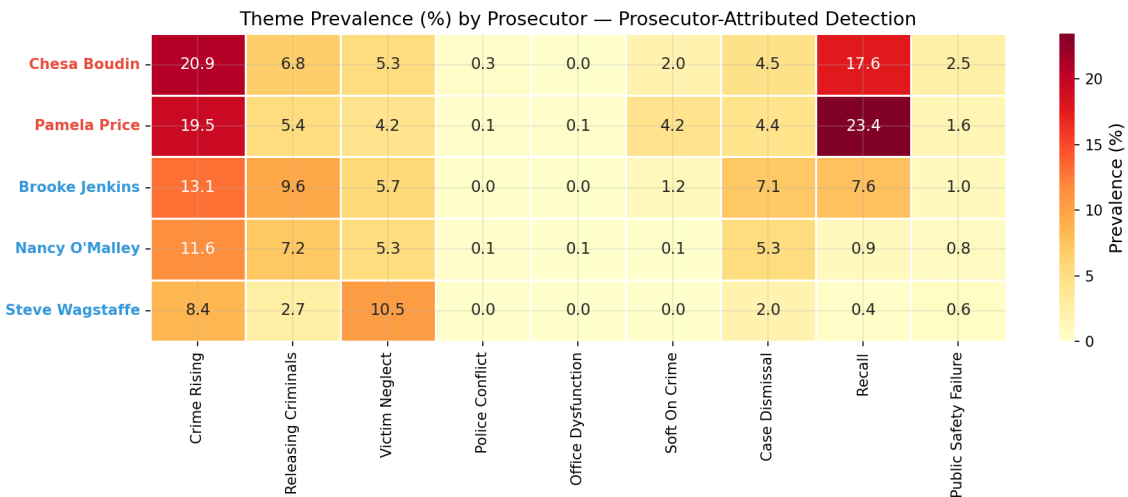


Figure 14: Theme prevalence (%) by individual prosecutor. Rows are colored by ideology (red = progressive, blue = traditional). Darker cells indicate higher prevalence.

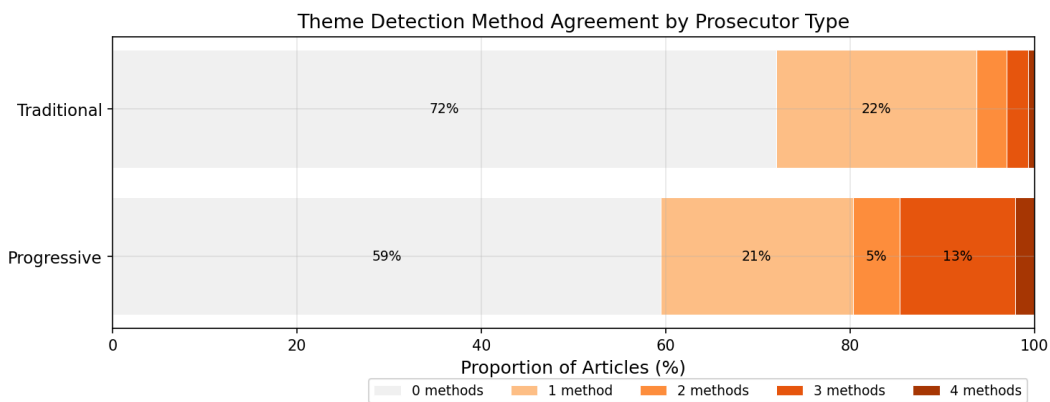


Figure 15: Distribution of detection method agreement by prosecutor type. Segments show the proportion of articles detected by 0–4 independent methods.

Appendix D Illustrative Examples of Identified Bias by Method

This appendix presents concrete examples from the corpus to illustrate what each method detects. Examples are drawn from articles at the extremes of each method’s score distribution and are intended to make the abstract measurement pipeline tangible.

Method A: Aspect-Based Sentiment Analysis

Method A extracts three-sentence context windows around prosecutor mentions. The following window scored $A = -0.932$ (*San Francisco Chronicle*, about Pamela Price):

“Don’t blame Alameda DA Price for crime. The pandemic changed everything, and we failed low-opportunity communities, especially our youth; crime in our cities has climbed in some areas and it is frightening.”

Although the article’s argument is *defensive* of Price, the RoBERTa model responds to the emotional valence of surrounding language, including “crime,” “climbed,” and “frightening,” rather than the argumentative intent. By contrast, a reader letter produced $A = +0.955$: “I am so thankful for Alameda County District Attorney Pamela Price. She has taken on an enormous job and she is doing it well.”

Method B: Zero-Shot Stance Classification

A paragraph from kron4.com received a critical stance score of $B = -0.996$:

“[The letter] cites failed leadership, the movement to defund the police and the district attorney’s unwillingness to charge and prosecute people who commit life threatening crimes.”

BART-MNLI classifies this as criticizing the prosecutor’s handling of crime with near-certainty. The divergence between Methods A and B ($d = 0.047$ for sentiment vs. $d = -0.337$ for stance)

is visible in individual examples: an article can score near zero on sentiment while scoring highly critical on stance.

Method C: Enhanced Keyword Analysis

An article about Nancy O'Malley (berkeleyside.org, 2021-07-18, "Alameda County District Attorney Nancy O'Malley is stepping aside") triggered two themes simultaneously: **releasing criminals** ("Boudin has **released** a person accused of a crime") and **recall** ("announced a **recall campaign** against O'Malley"). The keywords capture thematic framing rather than sentiment: "released" is emotionally neutral but thematically codes the releasing-criminals narrative. The same article also illustrates the dictionary's brittleness. It contains the phrase "very light sentences or no sentences at all" within a prosecutor mention window, but the soft-on-crime theme does not fire, because the dictionary entry is the singular "light sentence" and matching is word-boundary anchored. Dictionary methods of this kind miss surface variants they do not enumerate, which is one reason Method C carries the second-lowest composite weight.

Method D: Document-Level Sentiment

Because crime reporting is inherently negative, Method D reveals a paradoxical pattern: articles about traditional prosecutors are slightly *more* negative in overall tone ($d = +0.049$, $p = .005$). This underscores why prosecutor-specific methods are necessary.

Media Framing Analysis

The framing analysis classifies paragraphs (including standalone headlines) into five narrative frames. Illustrative examples include:

An earlier version of this table used "As new D.A., Jenkins vows to be tougher than Boudin" (*San Francisco Chronicle*, 2022-07-10) as the reform example. That article does score high on the reform frame (0.974), but its dominant frame is accountability (0.992), so it did not belong

Table 10: Illustrative examples of each dominant media frame (Appendix D). Each row is an article whose `dominant_frame` is the frame named in the first column; the score is that article’s probability on that frame, from `output/05_frames.parquet`. All five are zero-shot-classified rather than keyword-fallback articles.

Frame	Example Headline	Prosecutor	Score
Accountability	“SF DA Boudin Blamed by Some for Release of Parolee. . .”	Boudin	1.00
Conflict	“Jenkins blames Boudin in decision to drop case. . .”	Jenkins	1.00
Human interest	“Family of woman killed in S.F. New Year’s hit and run. . .”	Boudin	1.00
Reform	“Contrasting front-runners for D.A. stake out stances. . .”	Boudin	1.00
Consequences	“Price’s office lags on providing charging data”	Price	0.99

in a table of dominant-frame exemplars and has been replaced by a genuinely reform-dominant article (*San Francisco Chronicle*, 2019-11-01, `frame_reform` = 0.998). The substitution is a reminder that the five frame probabilities are estimated in multi-label mode and are not mutually exclusive: contested-election coverage routinely scores near ceiling on several frames at once, and the dominant-frame assignment resolves that by `argmax` rather than by exclusivity.

Group-level dominant-frame rates for these five frames are in Table 4; the pooling caveat in the Figure 8 caption applies to any per-prosecutor reading of frame scores.

Prosecutor-Attributed Theme Detection

The theme attribution method ($d = -0.418$) requires anti-prosecutor themes to be explicitly linked to a named prosecutor through compound linguistic patterns, distinguishing it from the proximity-based keyword method (Method C). An article titled “Boudin Blunders: SF DA’s downfall leading up to recall” (*kron4.com*, 2022-06-08) received a theme attribution score of 17.19, with all four independent detection methods firing and five themes identified: crime-rising, soft-on-crime, case-dismissal, recall, and public-safety-failure. That score is near the top of the distribution but is not the maximum; the highest score in the corpus is 17.50, for a *San Francisco Chronicle* article of 2022-03-16 reporting recall polling on Boudin (“Boudin might be in trouble if poll holds”). By contrast, a routine crime article about a man accused of hitting a woman with a car (*San Mateo Daily Journal*, 2021-09-24) attributed to Wagstaffe received a theme score of 0.0, illustrating

the method’s specificity in not flagging ordinary crime coverage merely because a prosecutor is mentioned.

LLM-Based Bias Indicator Detection

A *Mission Local* article about District Attorney Jenkins’s handling of the Keita O’Neil case (2023-03-07) produced 22 bias indicators, including **loaded language** (“killer cops” as pejorative label; “swept under the rug of political expediency” as ideological framing), **source imbalance** (extended critical quotes from the victim’s family and their attorney with no counterbalance), and **missing context** (the prosecutorial rationale for dismissing the charges is never presented). Critically, a large share of the sampled articles (428 of 999, 43%) produce zero indicators: evidence-based critical reporting is not flagged as biased. The method also detects bias *favoring* prosecutors, supporting the finding that bias is not unidirectional.

Appendix E Leave-One-Prosecutor-Out Sensitivity

The pooled cross-sectional contrast compares five prosecutors in three counties, so it can be driven by any one of them. Table 11 refits the group comparison five times, each time dropping one prosecutor and retaining the other four. An earlier version of this appendix reported only the leave-Jenkins-out fit, which strengthens the effect; the full set does not support a uniform-robustness claim.

Table 11: Leave-one-prosecutor-out group comparisons. “% of full” is the composite d as a percentage of the full-sample composite d . Composite and keyword n ’s equal the article counts shown; stance n ’s are smaller because the stance score is missing for a subset of articles (full sample: $n = 5,186$ progressive, 4,719 traditional).

Sample	n_{Prog}	n_{Trad}	Composite d	% of full	Stance d	Keywords d
Full sample	6,585	6,242	−0.151	100%	−0.337	−0.203
Drop Boudin (Prog., SF)	802	6,242	−0.150	99%	−0.437	−0.210
Drop Price (Prog., Alameda)	5,783	6,242	−0.154	102%	−0.330	−0.207
Drop Jenkins (Trad., SF)	6,585	3,145	−0.233	154%	−0.467	−0.306
Drop O’Malley (Trad., Alameda)	6,585	4,745	−0.173	115%	−0.371	−0.160
Drop Wagstaffe (Trad., San Mateo)	6,585	4,594	−0.070	46%	−0.206	−0.162

Four of the five fits leave the composite between $d = -0.150$ and $d = -0.233$. The fifth does not: dropping Steve Wagstaffe cuts the composite to $d = -0.070$ ($p = 2.4 \times 10^{-4}$), 46% of the full-sample effect, and cuts stance from $d = -0.337$ to $d = -0.206$. The full leave-one-out range is $[-0.233, -0.070]$, and every fit remains negative and significant, so the sign and the direction of the contrast are robust while its magnitude is not.

The mechanism is Wagstaffe’s position in the design rather than anything about the measurement. His coverage is the most favorable of any prosecutor in the study on both the composite ($M = +0.040$, versus -0.016 for O’Malley, -0.017 for Jenkins, -0.029 for Price, and -0.031 for Boudin) and stance ($M = 0.473$, versus 0.278 and 0.279 for the other two traditional prosecutors and 0.147 and 0.189 for the progressives), and he contributes 1,648 articles, 26% of the traditional baseline. He is also the only prosecutor from San Mateo County, the only county in the study with no prosecutorial transition. The pooled cross-sectional contrast is therefore partly a comparison between counties rather than between prosecutor types, and it is partly dependent on a single untreated county.

Read against the within-county estimates, this is a coherence gain rather than a loss: the drop-Wagstaffe fit, the San Francisco pair, and the Alameda pair all land at $d \approx -0.07$ (Section 4.2).

One tension deserves explicit statement. San Mateo County is load-bearing in two places at once: Wagstaffe supplies the traditional-baseline articles whose removal halves the pooled cross-sectional effect, and San Mateo is also the untreated comparison series in the controlled segmented ITS (Appendix G). These are different estimands—the controlled ITS differences out San Mateo’s monthly *time trend* rather than comparing levels against it, so the pooled contrast and the controlled ITS are not the same test run twice—but the same 1,648 articles from a single county carry weight in both, and no second untreated county is available to check either result.

For completeness, the per-method detail of the leave-Jenkins-out fit is retained in Table 12, since Brooke Jenkins accounts for roughly half of the traditional baseline ($n = 3,097$ of $n = 6,242$ traditional articles) and this was the originally reported check ($n = 6,585$ progressive; $n = 3,145$ traditional).

Table 12: Per-method sensitivity to excluding Brooke Jenkins from the traditional baseline.

Outcome	Full sample (d, p)	Excluding Jenkins (d, p)
Composite	$d = -0.151, p = 1.07 \times 10^{-17}$	$d = -0.233, p = 2.94 \times 10^{-30}$
Stance	$d = -0.337, p = 4.90 \times 10^{-63}$	$d = -0.467, p = 4.42 \times 10^{-90}$
Keywords	$d = -0.203, p = 7.92 \times 10^{-31}$	$d = -0.306, p = 2.28 \times 10^{-67}$
Aspect sentiment	$d = 0.047, p = 0.018$	$d = 0.025, p = 0.288$
Document sentiment	$d = 0.049, p = 0.005$	$d = 0.022, p = 0.309$

Excluding Jenkins strengthens the composite, stance, and keyword differentials materially, while sentiment metrics remain small and comparatively weak. Taken alone this check suggests that the full-sample estimates are conservative with respect to evaluative dimensions, but Table 11 shows that the inference does not generalize: removing a different traditional prosecutor moves

the composite the other way, by a comparable amount.

For the complementary robustness check that removes fallback-attributed articles instead of a specific prosecutor baseline component, see Appendix F.

Appendix F Sample-Composition Sensitivity Analyses

In the attribution pipeline, fallback assignments occur when an article contains generic DA references (e.g., “the district attorney”) but no named prosecutor mention; in these cases, the article is assigned by publication county and article date. This mechanism expands coverage but can introduce measurement noise when outlets report cross-county stories.

Using the article-level analysis sample from Section 4 ($n = 12,827$), fallback assignment accounts for 6,651 articles (51.9%). Excluding these fallback-attributed cases leaves 6,176 directly attributed articles (named-mention based), with $n = 3,322$ progressive and $n = 2,854$ traditional.

Before the estimates: this exclusion does not define a new subsample. The articles retained here are, up to 29 cases out of 12,827, the same articles the framing layer classifies with the zero-shot model rather than the keyword fallback (Jaccard = 0.996 between the two indicators; Section 3.4). The exclude-fallback composite sensitivity below and the model-classified framing analysis in Section 4.6 are therefore the same subsample viewed through two instruments, and their agreement should be read as one fact about that subsample rather than as two independent confirmations.

Table 13: Composite group comparison: full sample vs. sample-composition sensitivities.

Sample	n_{Prog}	M_{Prog}	n_{Trad}	M_{Trad}	d	p_{Welch}
Full attributed sample	6,585	-0.0304	6,242	-0.0020	-0.151	1.07×10^{-17}
Exclude fallback-assigned	3,322	-0.0385	2,854	+0.0213	-0.294	3.18×10^{-31}
Prosecutor-focused subsample	2,209	-0.0432	1,790	+0.0052	-0.260	1.1×10^{-16}
Mention-dominance subsample	2,043	-0.0429	1,738	+0.0068	-0.267	1.3×10^{-16}
Tenure-period only	6,314	-0.0378	6,234	-0.0020	-0.192	8.8×10^{-27}

Table 13 reports all five estimates. The no-fallback estimate is approximately double the full-sample effect size in magnitude. The bootstrap mean difference (progressive minus traditional) is -0.0598 with 95% CI $[-0.0700, -0.0498]$, confirming that the progressive group remains significantly more negative after removing all fallback assignments. Substantively, this indicates that fallback attribution is conservative in this corpus: it attenuates the ideology differential rather than driving it.

Three further sample-composition sensitivities, motivated by the human validation audit (Section 4.9), probe attribution quality from complementary directions. First, a *prosecutor-focused* subsample retains an article only if it mentions the primary prosecutor at least twice by name or mentions the prosecutor in the headline. The rule’s grading against the research-assistant-coded benchmarks, and the script that regenerates it, are reported in Section 4.9. Its residual failures are articles that name the attributed prosecutor repeatedly while actually being about another prosecutor, so the subsample retains some peripheral coverage. The rule keeps 3,999 of 12,827 articles, and the composite differential increases to $d = -0.260$ ($p = 1.1 \times 10^{-16}$). Like the fallback exclusion, this indicates that articles genuinely centered on the attributed prosecutor exhibit a larger differential than the full attributed sample.

Second, a *mention-dominance* subsample tightens the same relevance logic to address that residual

failure mode. The rule requires, in addition to the mentions-or-headline condition, that the primary prosecutor be named more often than all other prosecutors combined, counting both the study’s other four and prominent non-study prosecutors (Gascón, Harris, Rosen, Spitzer, Chisholm, Krasner, Mosby, and Foxx). It targets the residual failure mode identified in Section 4.9 and improves benchmark performance there. The rule keeps 3,781 of 12,827 articles, and the composite differential is $d = -0.267$ ($p = 1.3 \times 10^{-16}$). That improvement is graded in-sample (Section 4.9), so it is an upper bound on what the added condition buys rather than a validated gain over the simpler rule, and out-of-sample confirmation with the second blinded coder is pending.

The three relevance-based restrictions point the same way. Excluding fallback assignments ($d = -0.294$), restricting to prosecutor-focused articles ($d = -0.260$), and applying the mention-dominance rule ($d = -0.267$) each roughly double the full-sample effect ($d = -0.151$) in the same direction. It would overstate the evidence to call this convergence, because the three subsamples are nested rather than merely overlapping: mention-dominance (3,781 articles) is by construction a subset of prosecutor-focused (3,999), which is in turn a near-subset of the non-fallback sample (3,991 of 3,999 articles), since requiring two named mentions implies at least one. The three estimates are therefore one restriction applied at three levels of severity, and the agreement among them shows that the effect is monotone in relevance, not that three independent tests concur. What that monotonicity does establish is the pattern expected if peripheral and misattributed articles contribute noise centered near zero rather than generating the differential: restricting the corpus to coverage genuinely about the attributed prosecutor strengthens rather than weakens the contrast. This convergence concerns *relevance* restrictions only and should not be generalized to composition: the pooled estimate is sensitive to which prosecutors compose the two groups, and one leave-one-out fit halves it (Appendix E).

Third, a *tenure-period-only* sample excludes the 279 articles published before the attributed prosecutor took office, most of which are campaign coverage (208 concern Boudin’s 2019 campaign). The composite differential is essentially unchanged and slightly strengthened ($d = -0.192$, $p =$

8.8×10^{-27}), indicating that pre-tenure campaign coverage does not drive the main result.

Appendix G Segmented Interrupted Time-Series Robustness

The main text reports a concise segmented ITS summary for temporal dynamics by method. This appendix provides the full model specification, coefficient table, and plots for those temporal analyses:

$$Y_t = \beta_0 + \beta_1 t + \beta_2 \text{Post}_t + \beta_3 \text{TimeAfter}_t + \varepsilon_t$$

where β_2 captures an immediate level shift at transition and β_3 captures post-transition slope change. The transition month itself is coded as the first post-transition month ($\text{TimeAfter} = 0$ in that month), so β_2 is the level change at the first month of the new prosecutor’s tenure. Models are weighted by monthly article counts and use Newey–West (HAC) robust standard errors.

Table 14: Segmented ITS results (monthly, weighted, HAC-robust).

Transition	Outcome	Level shift β_2	Slope change β_3	12-month net effect
SF (Boudin → Jenkins)	Composite	+0.031 ($p = 0.070$)	+0.0031 ($p = 0.031$)	+0.069 ($p < .001$)
SF (Boudin → Jenkins)	Stance	+0.067 ($p = 0.157$)	+0.0125 ($p < .001$)	+0.216 ($p < .001$)
SF (Boudin → Jenkins)	Keywords	+0.002 ($p = 0.689$)	+0.0019 ($p < .001$)	+0.025 ($p = 0.010$)
Alameda (O’Malley → Price)	Composite	−0.067 ($p = 0.035$)	−0.0000 ($p = 0.993$)	−0.068 ($p = 0.030$)
Alameda (O’Malley → Price)	Stance	−0.238 ($p < .001$)	−0.0045 ($p = 0.569$)	−0.292 ($p < .001$)
Alameda (O’Malley → Price)	Keywords	+0.003 ($p = 0.265$)	−0.0020 ($p < .001$)	−0.021 ($p < .001$)

Figure 16 visualizes the same segmented models for the three most policy-relevant outcomes (composite, stance, keywords), plotting observed monthly means and fitted segmented trends around each transition date.

Segmented ITS Robustness: Monthly Outcomes and Fitted Pre/Post Trends

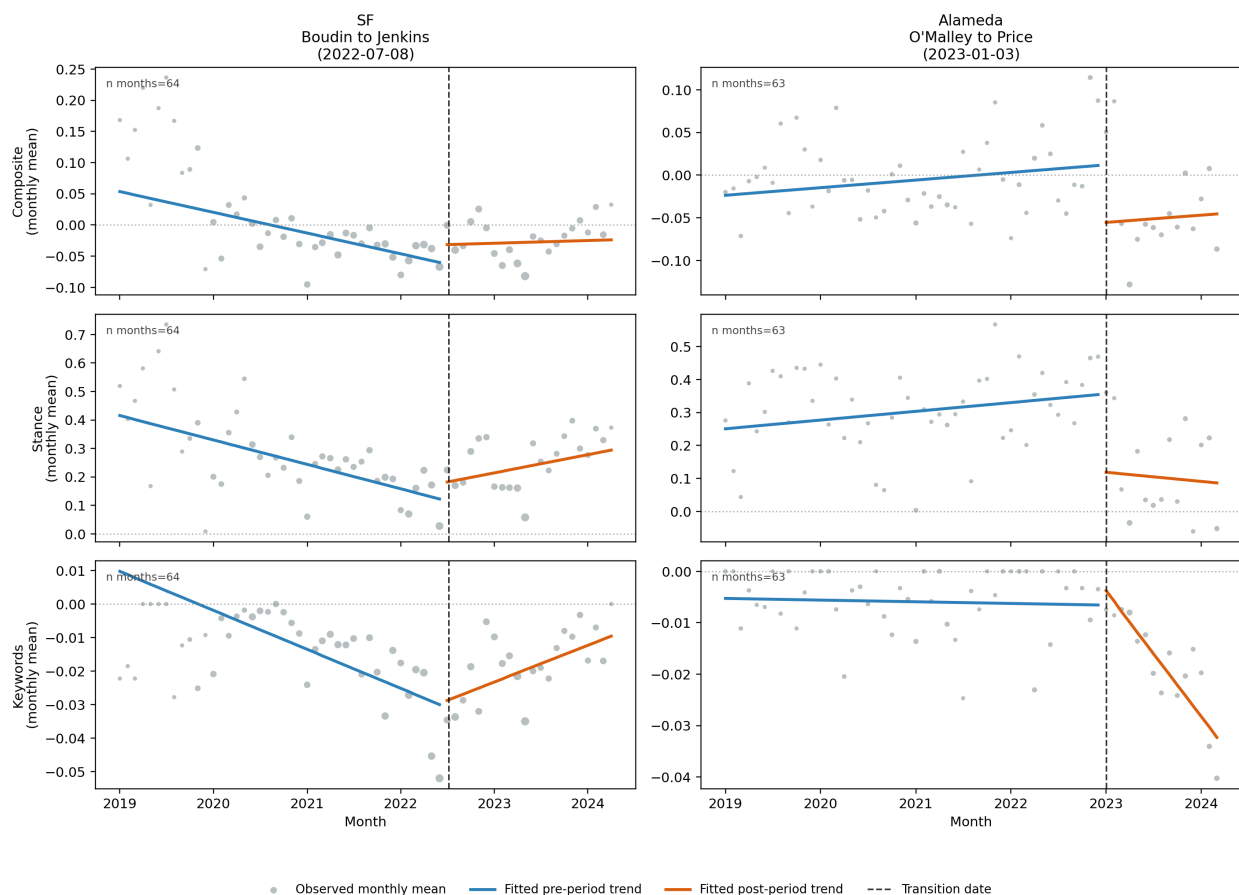


Figure 16: Segmented ITS robustness plots. Points are monthly outcome means (marker size proportional to monthly article count). Colored lines are fitted segmented trends before and after transition; dashed vertical lines mark transition dates.

These segmented models sharpen interpretation by separating immediate transition shifts from post-transition trend changes; the coefficients are in Table 14 and are summarized in Section 4. Across both transitions, stance remains the strongest and most consistent evaluative signal.

Because the segmented models involve a 30-test family (two transitions $\times$ five outcomes $\times$ three quantities: level change, slope change, and 12-month horizon effect), Benjamini–Hochberg adjusted p -values are computed across the full family (replication output `12_segmented_its_table.csv`). The adjustment changes several conclusions. The San Francisco composite level shift is not significant even before correction ($p = .070$); under the earlier, time-asymmetric attribution filter this co-

efficient had appeared nominally significant, and its disappearance under the corrected attribution is expected, because the 126 removed post-tenure false positives were concentrated precisely in the post-transition window (see Section 3.2). The San Francisco composite slope change ($p = .031$, $p_{BH} = .078$), the Alameda composite level shift ($p = .035$, $p_{BH} = .080$), and the Alameda composite 12-month horizon effect ($p = .030$, $p_{BH} = .078$) are no longer significant after correction and should be treated as suggestive rather than confirmed. The effects that survive correction are concentrated in the evaluative components. In San Francisco: the stance slope change ($p_{BH} = .002$) and 12-month effect ($p_{BH} < .001$), the keyword slope change ($p_{BH} = .001$) and 12-month effect ($p_{BH} = .032$), and the composite 12-month horizon effect ($p_{BH} = .002$). In Alameda: the stance level shift ($p_{BH} = .002$) and 12-month effect ($p_{BH} < .001$), and the keyword slope change and 12-month effect (both $p_{BH} < .001$). No sentiment-outcome test survives correction. The family-wise-robust conclusion from the ITS analysis is therefore about the evaluative channels, not the composite: transition-linked shifts in stance and keyword salience are reliable, while the composite-level step changes are not separately established.

As a further check against region-wide trends, I re-estimate the segmented models on a controlled difference series: each treated county's monthly outcome minus the contemporaneous monthly outcome for San Mateo County (Steve Wagstaffe), an untreated comparison county that experienced no prosecutor transition over the study period. In this controlled specification, the San Francisco composite level shift is no longer distinguishable from contemporaneous county-general trends ($\beta_2 = +0.028$, $p = .26$; the controlled San Francisco stance level shift is likewise non-significant, $p = .096$), suggesting that part of the San Francisco pattern reflects Bay Area-wide movement in crime-news coverage. The Alameda composite level shift, by contrast, survives the control and sharpens ($\beta_2 = -0.087$, $p = .003$), with the stance outcome showing the same pattern ($\beta_2 = -0.242$, $p < .001$); the step increase in negative Alameda coverage at the Price transition therefore cannot be attributed to region-wide shifts in coverage tone. The controlled contrast is a separate, pre-specified single comparison per county rather than part of the 30-test uncontrolled family, so it is not folded into the Benjamini–Hochberg adjustment above; read together, the two

analyses agree that the Alameda transition effect is the more reliable one, carried by the stance channel (which survives family-wise correction) and confirmed against the untreated county, while the San Francisco composite-level results are suggestive under both lenses. These robustness analyses improve temporal identification but should still be interpreted descriptively: the controlled contrast relies on a single comparison county and cannot rule out county-specific confounds. The same 1,648 Wagstaffe articles also constitute 26% of the traditional baseline in the pooled cross-sectional contrast, so one county is load-bearing in two places; the two uses are different estimands, but neither can be checked against a second untreated county (Appendix [E](#)).